Queues Crash Course

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Neural Autonomic Transport System

Not Another Topic System

Natively Asynchronous Transport System

NATS

- Architecture overview
- Key approaches
  - Pub/Sub
  - Req/Res
  - Queues
- Clustering
- Stream architecture: JetStream & Consumers

NATS Essentials

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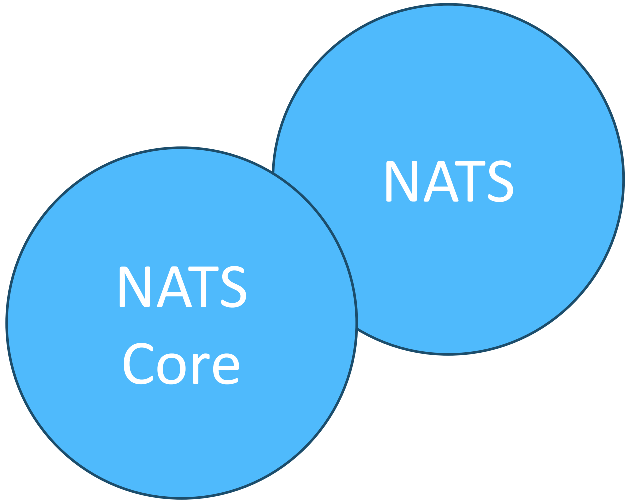
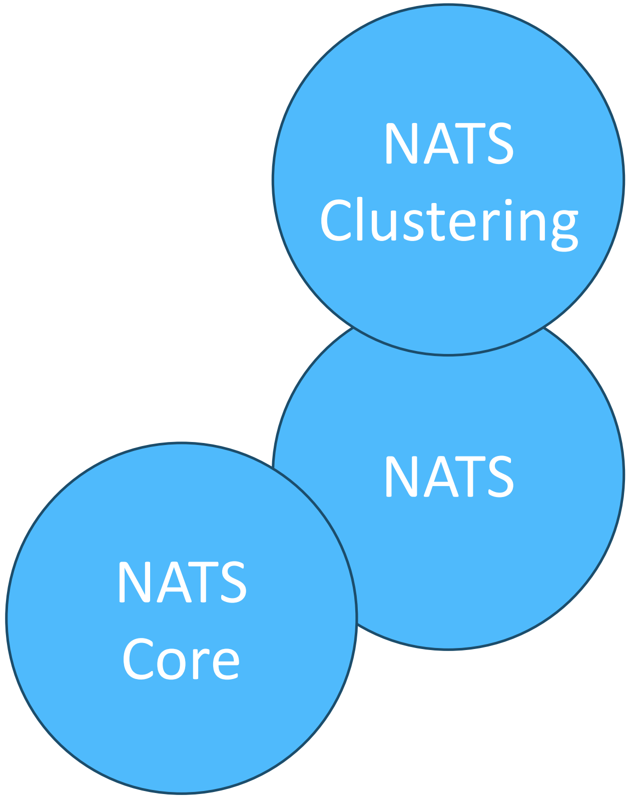
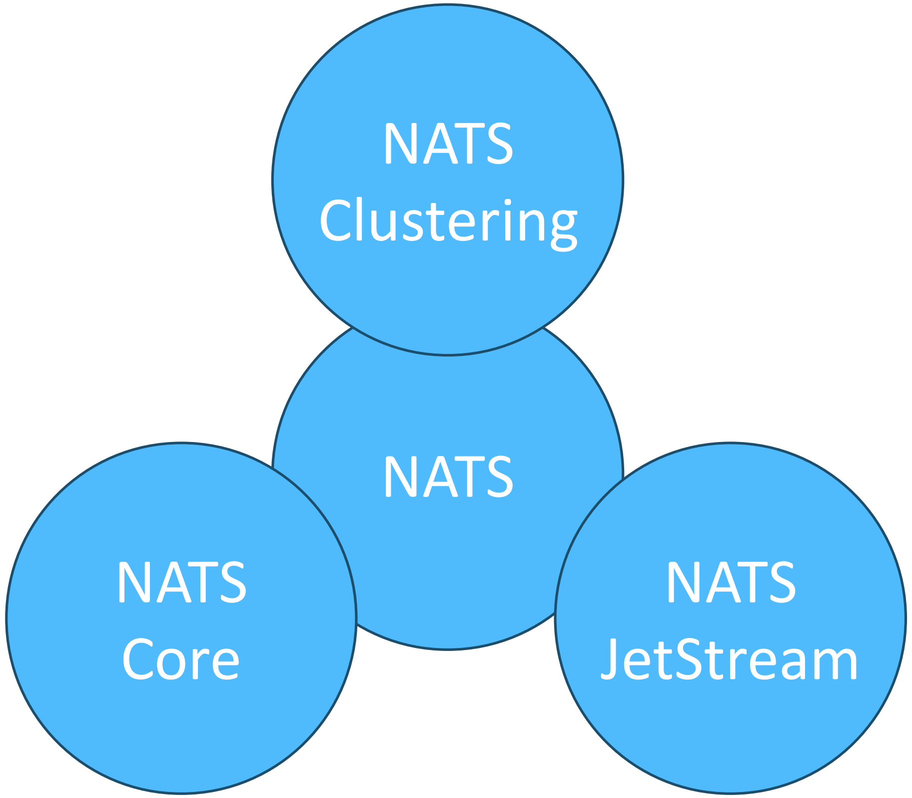


NATS Essentials



NATS Essentials



NATS Core: Messaging

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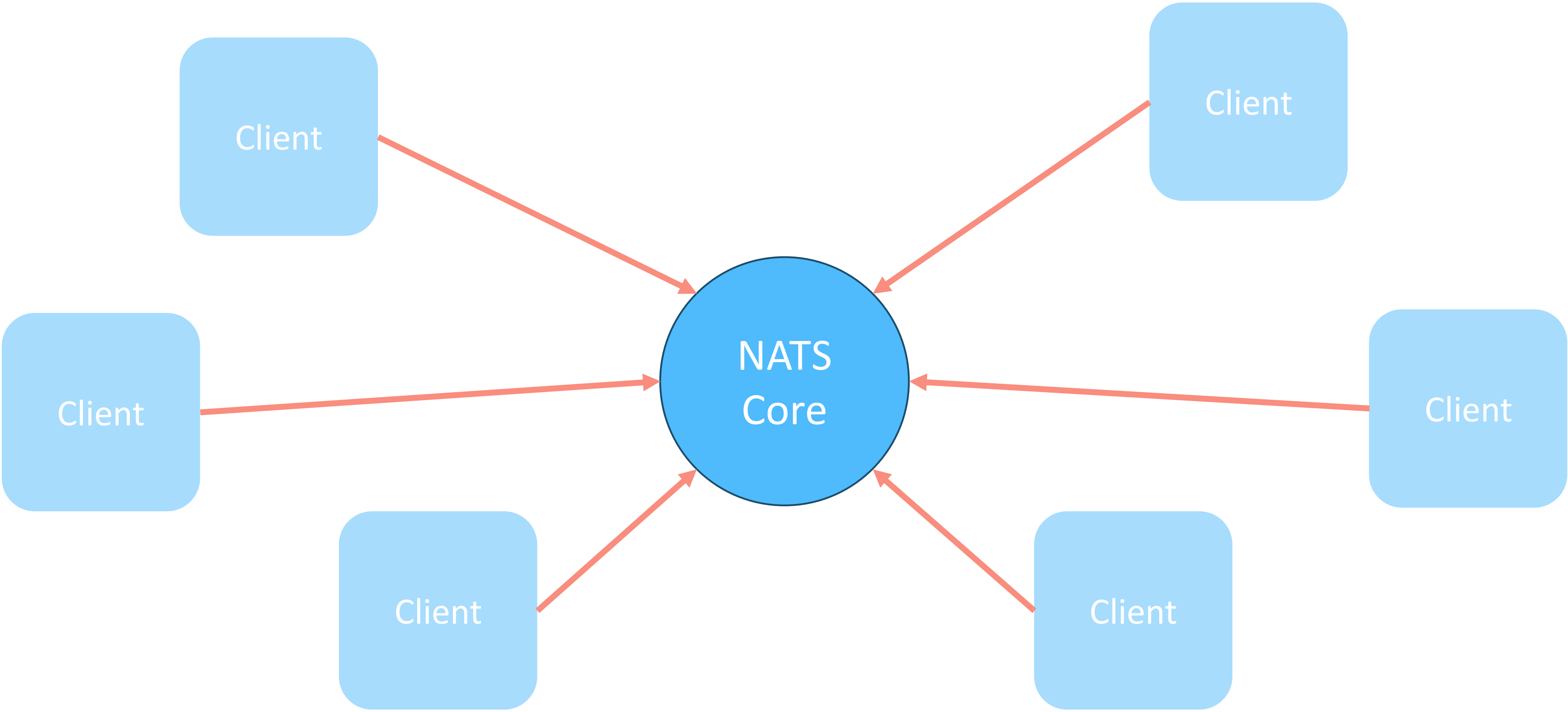

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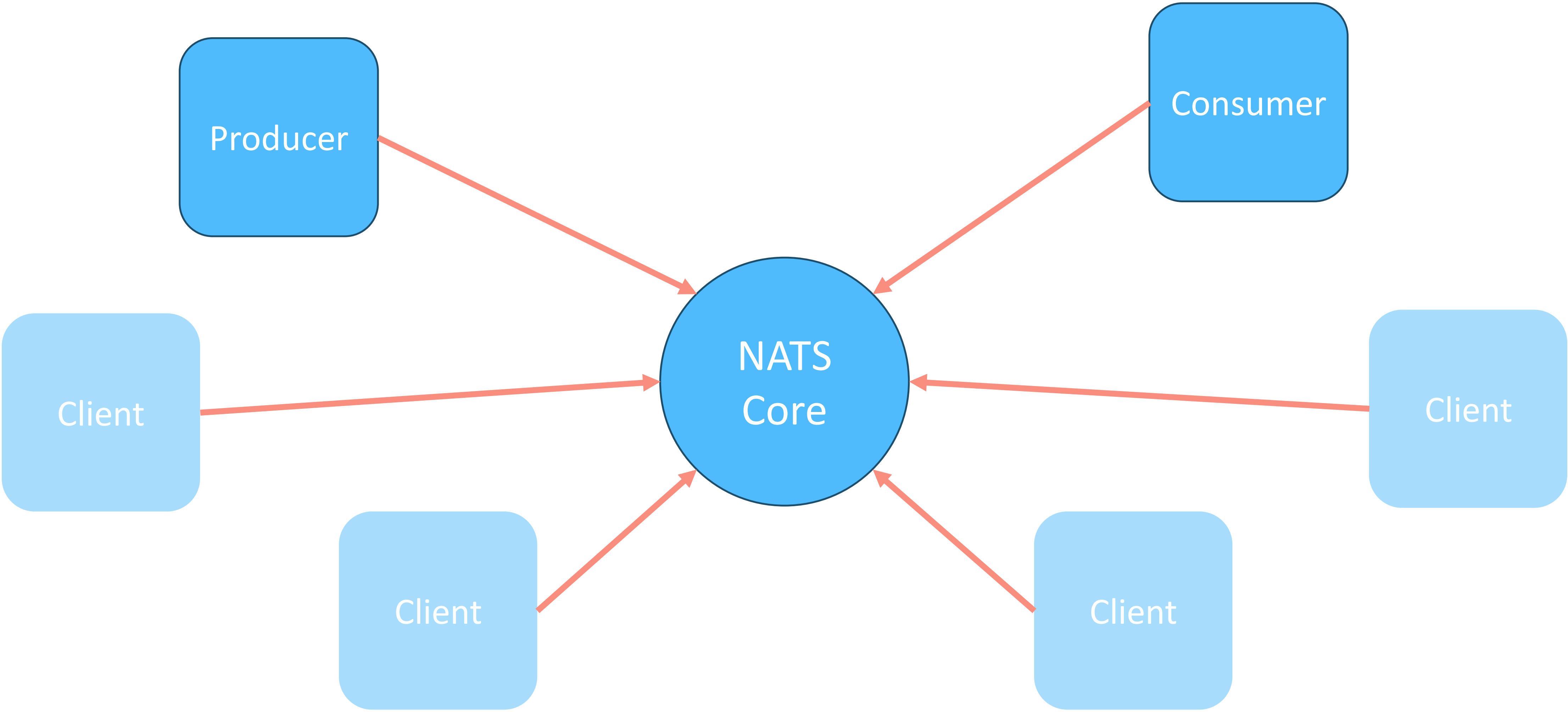

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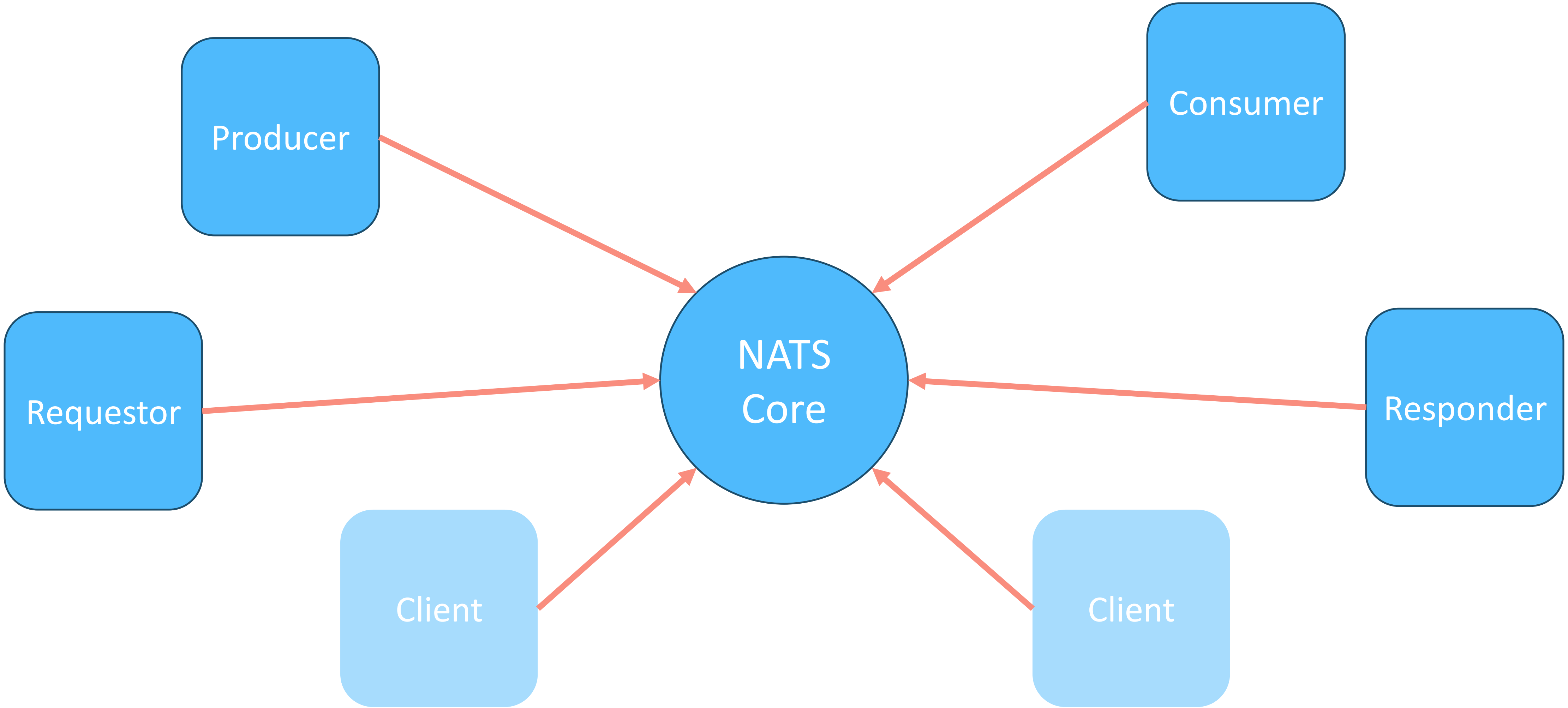

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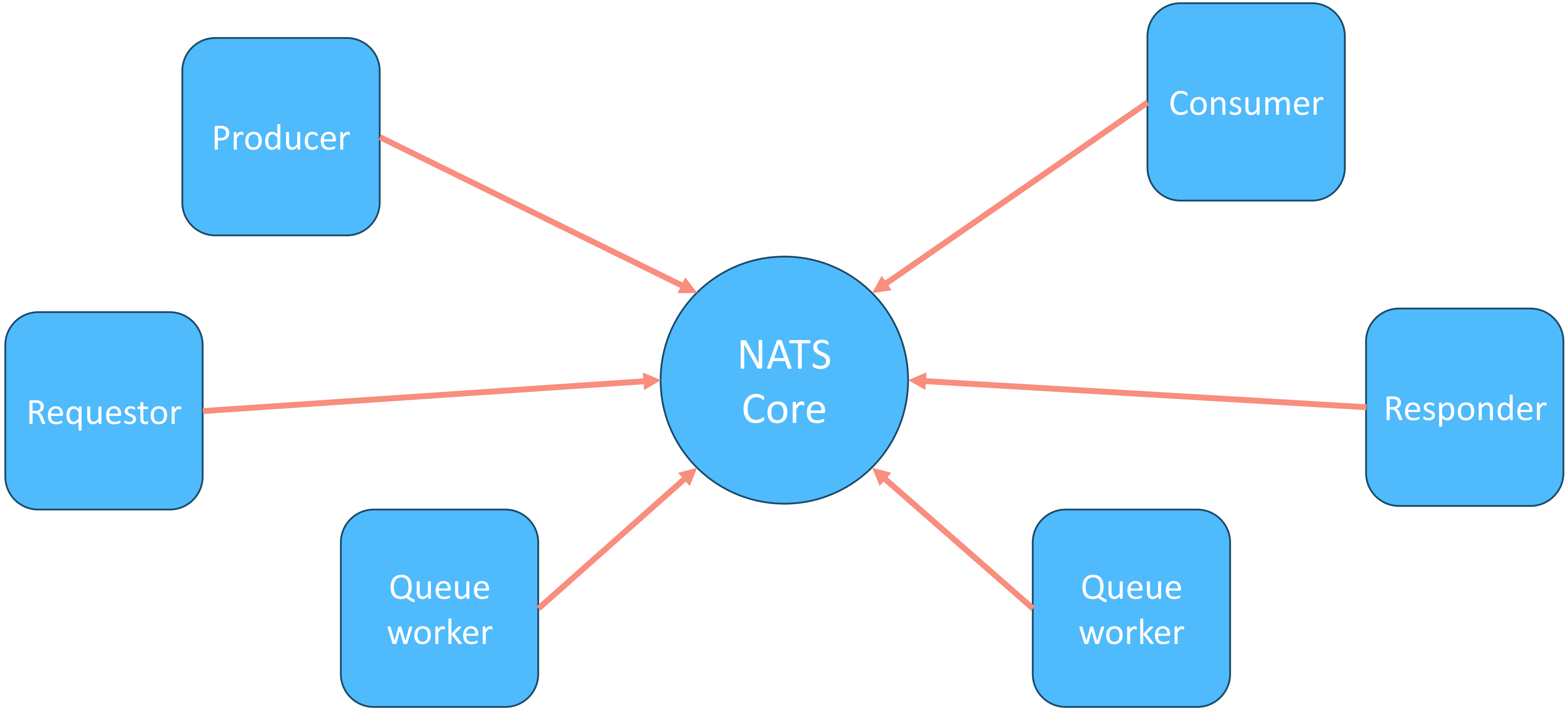
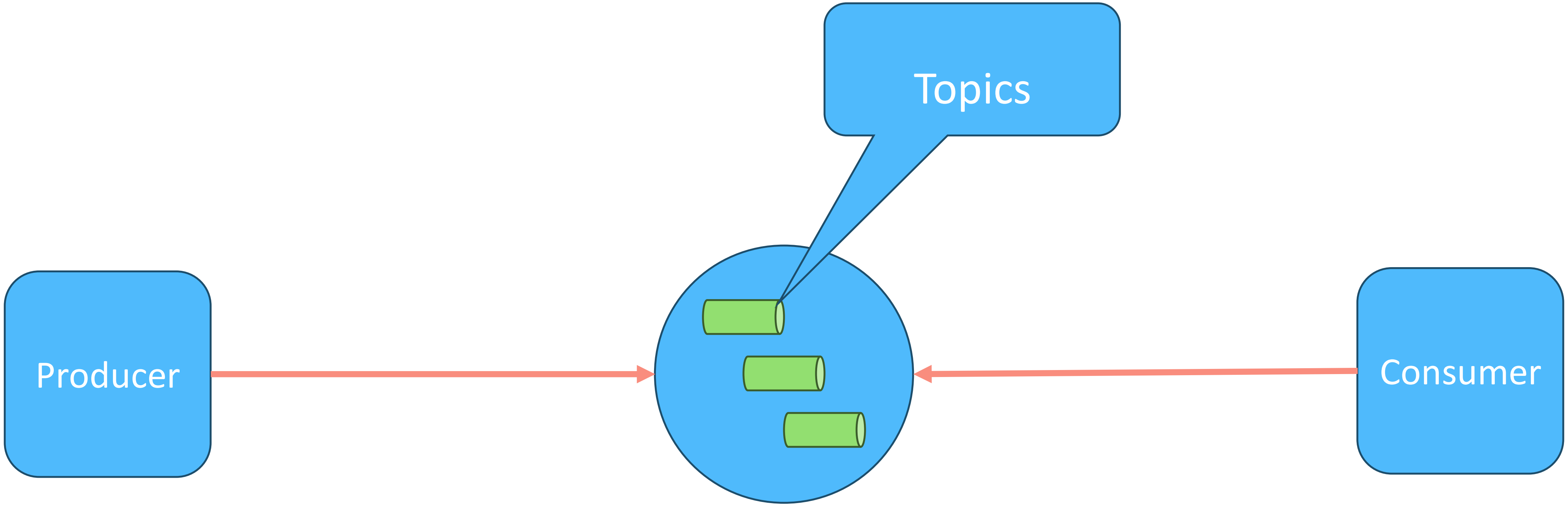


NATS Core: Pub/Sub



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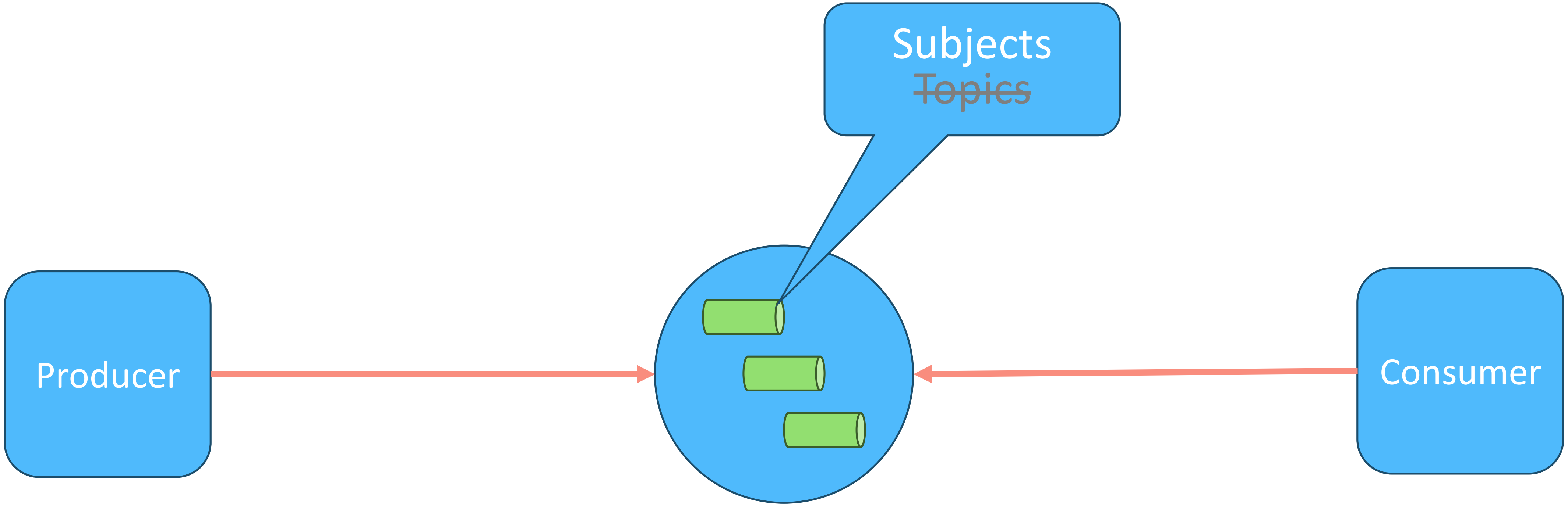

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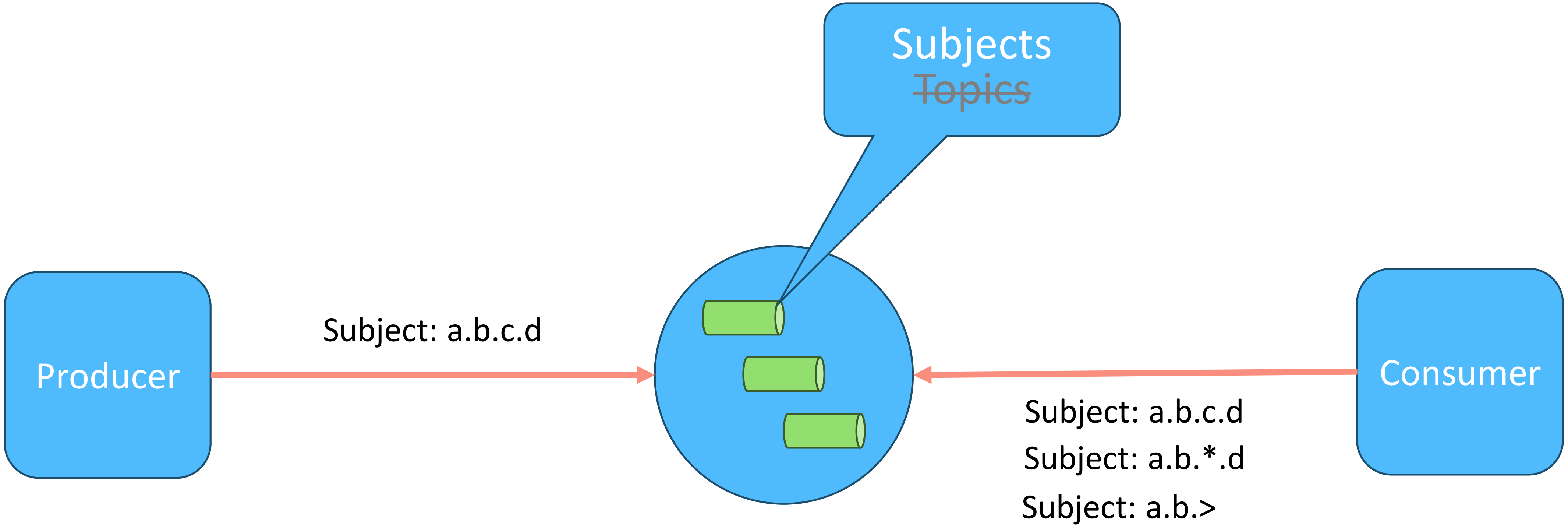

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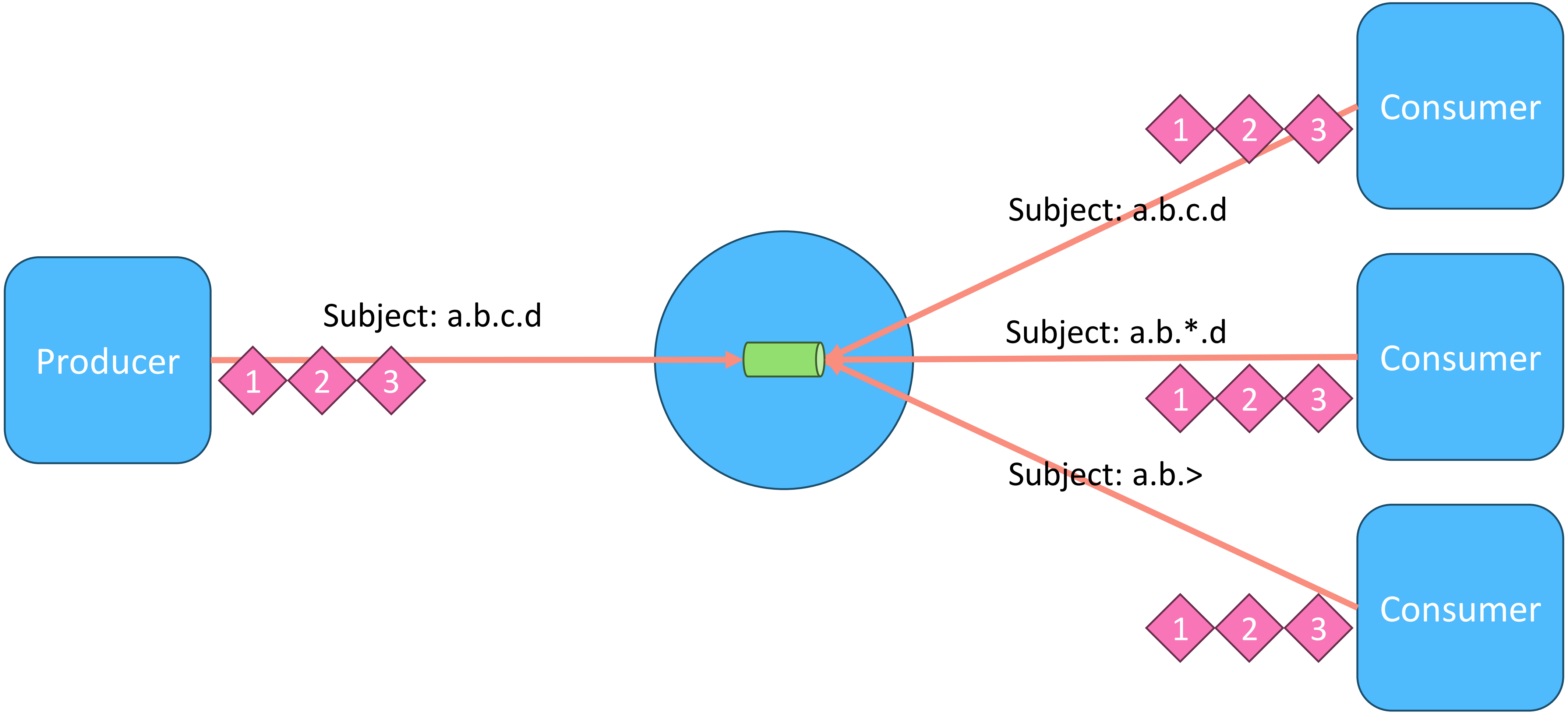
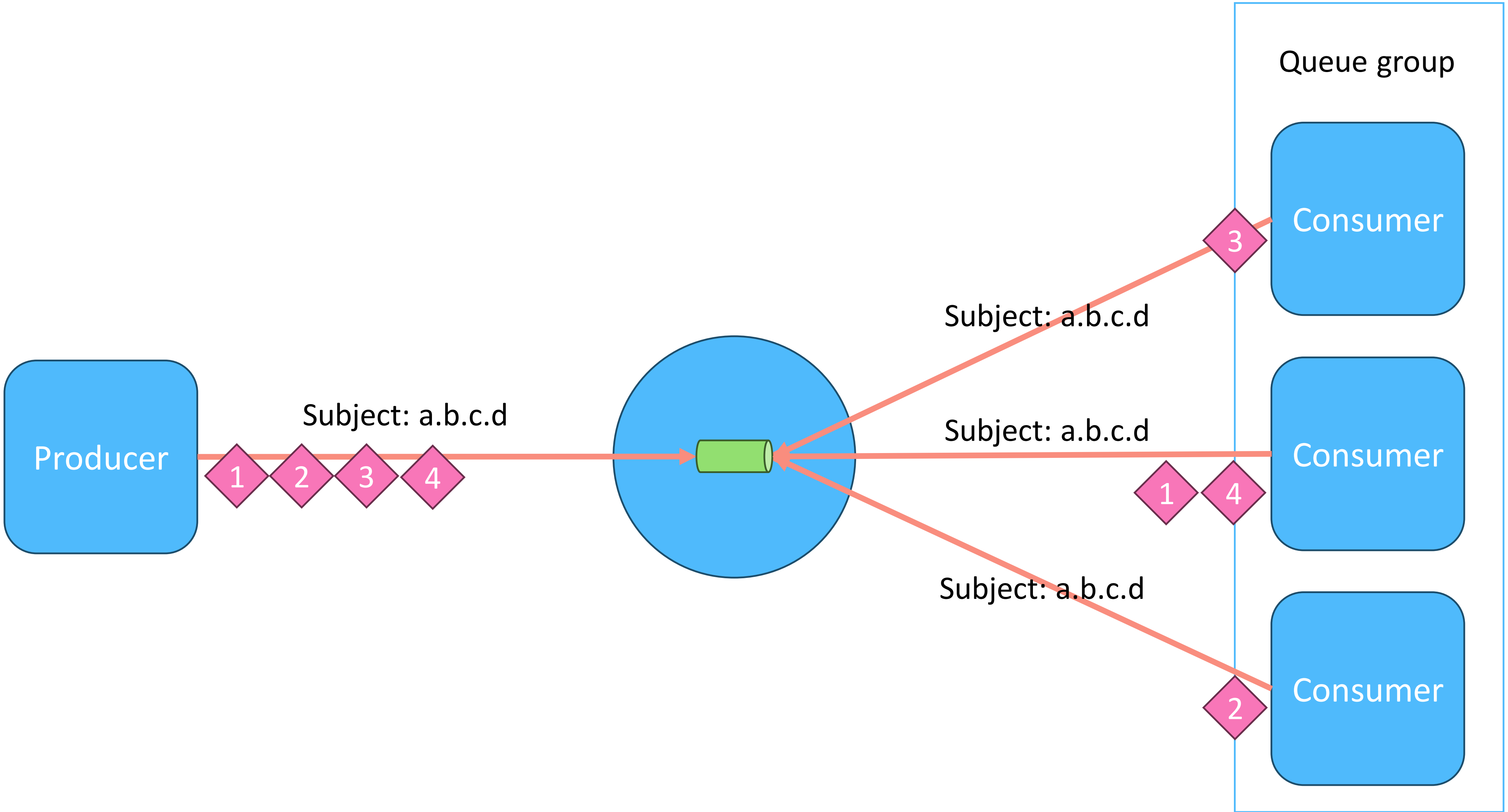
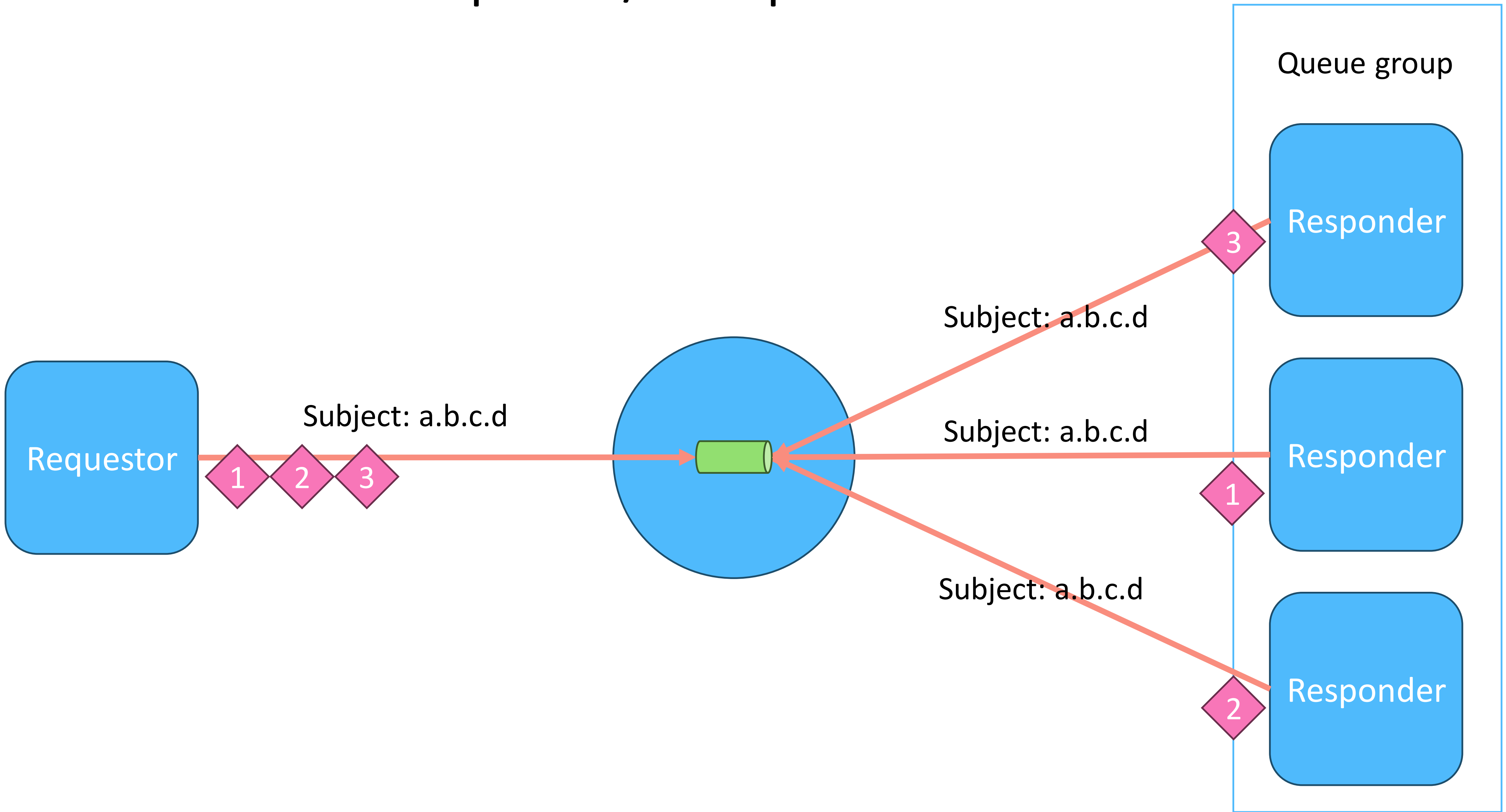


NATS Core: Queue



NATS Core: Request/Response



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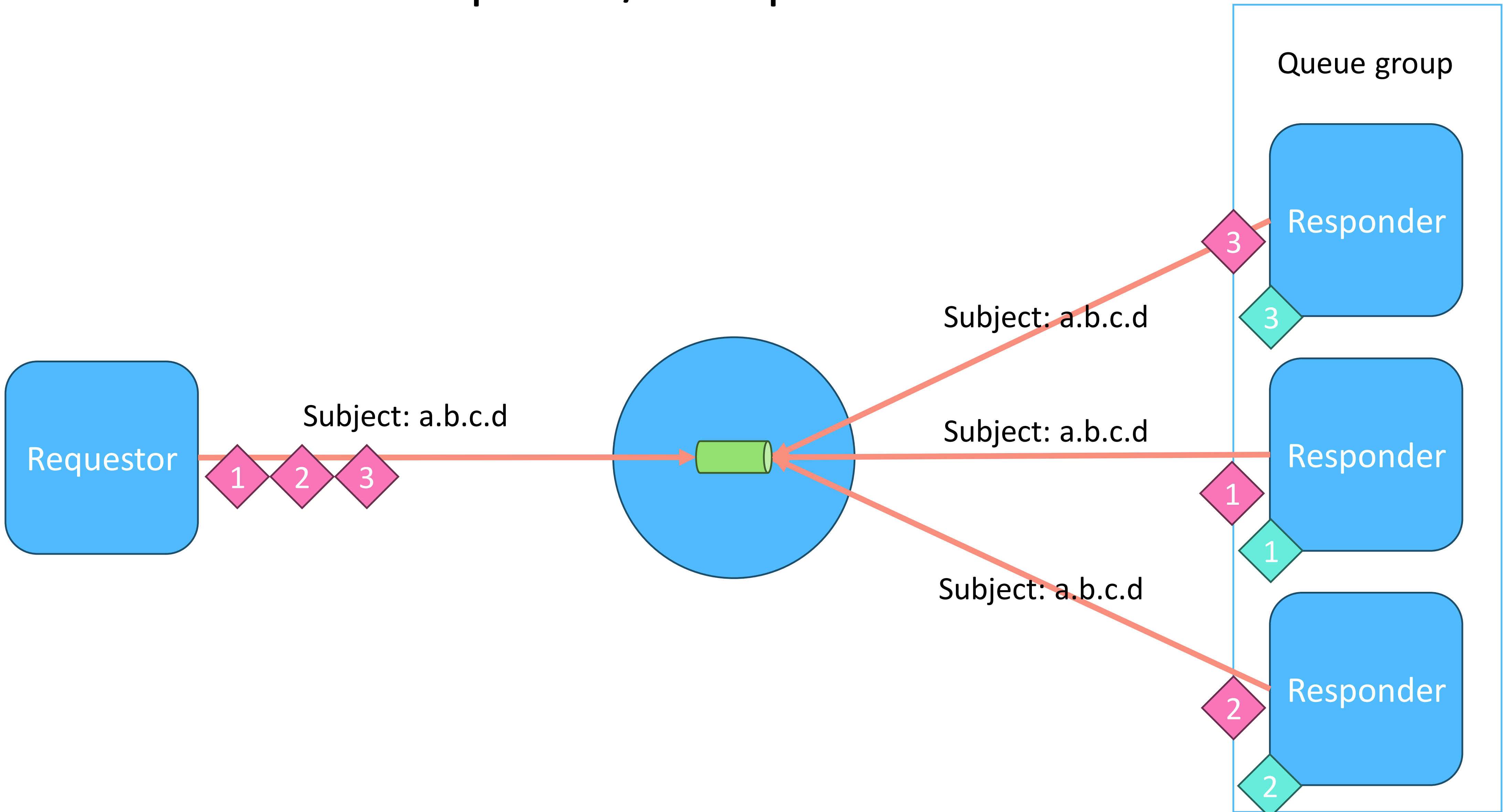

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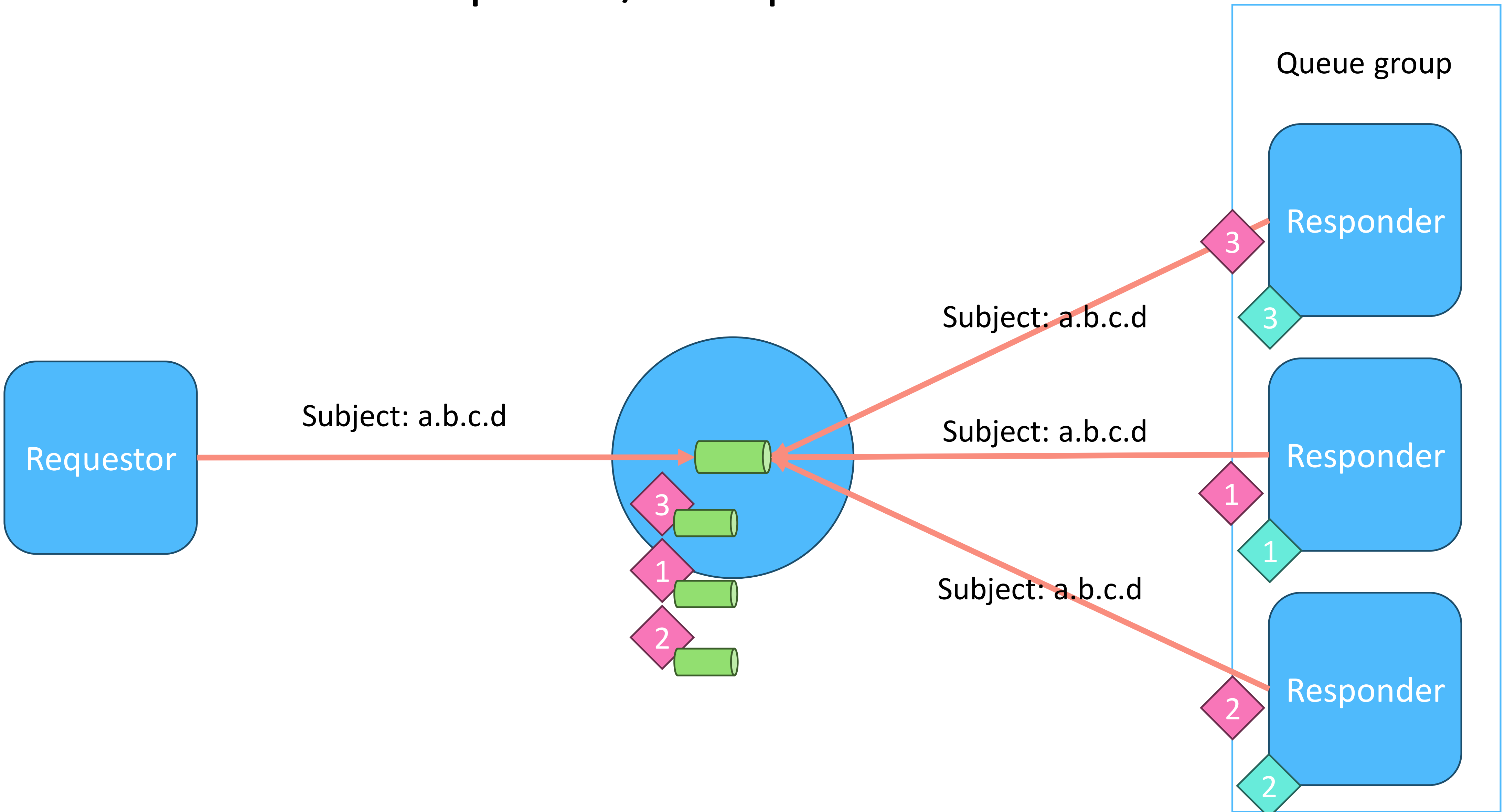

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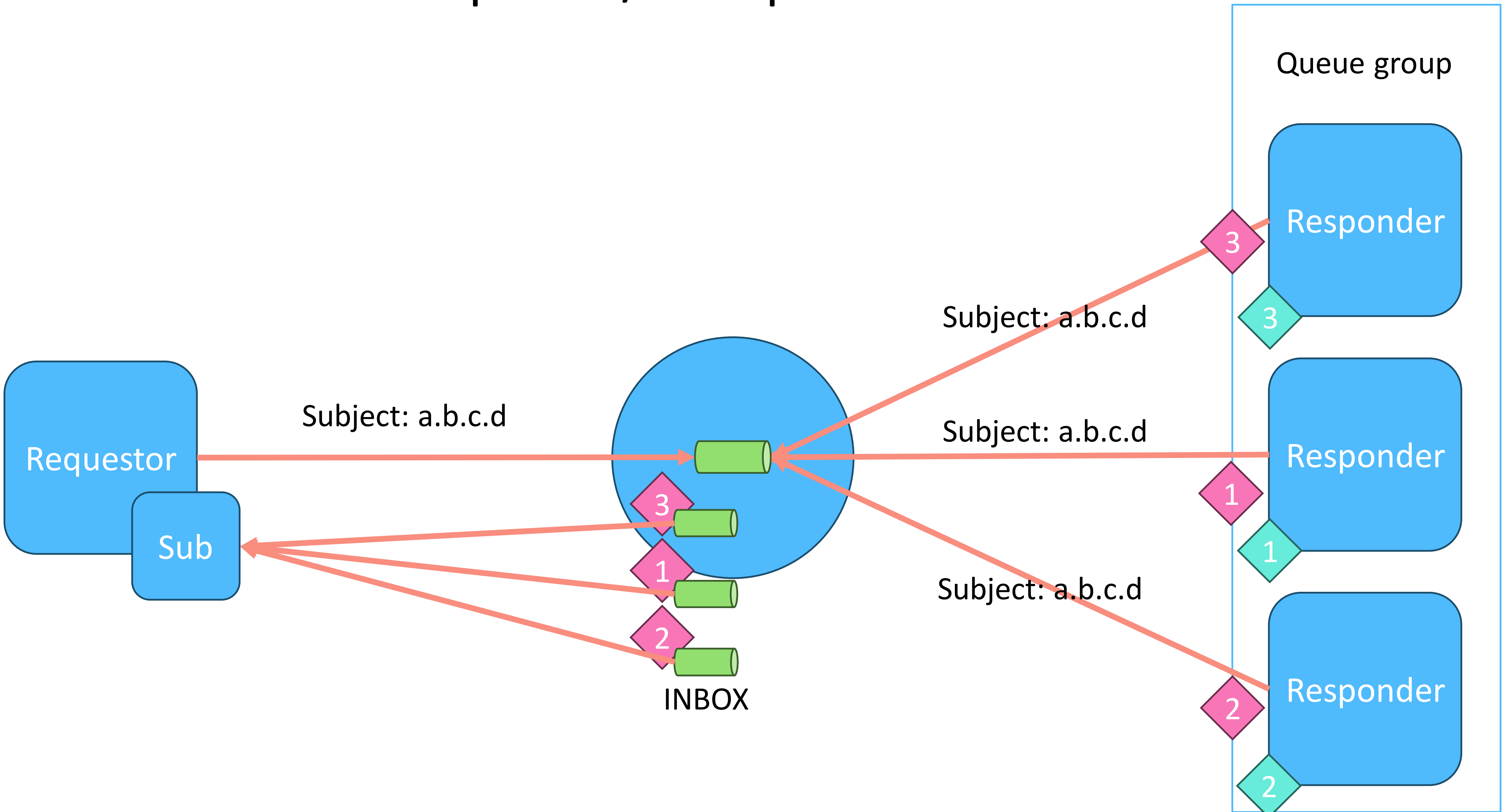

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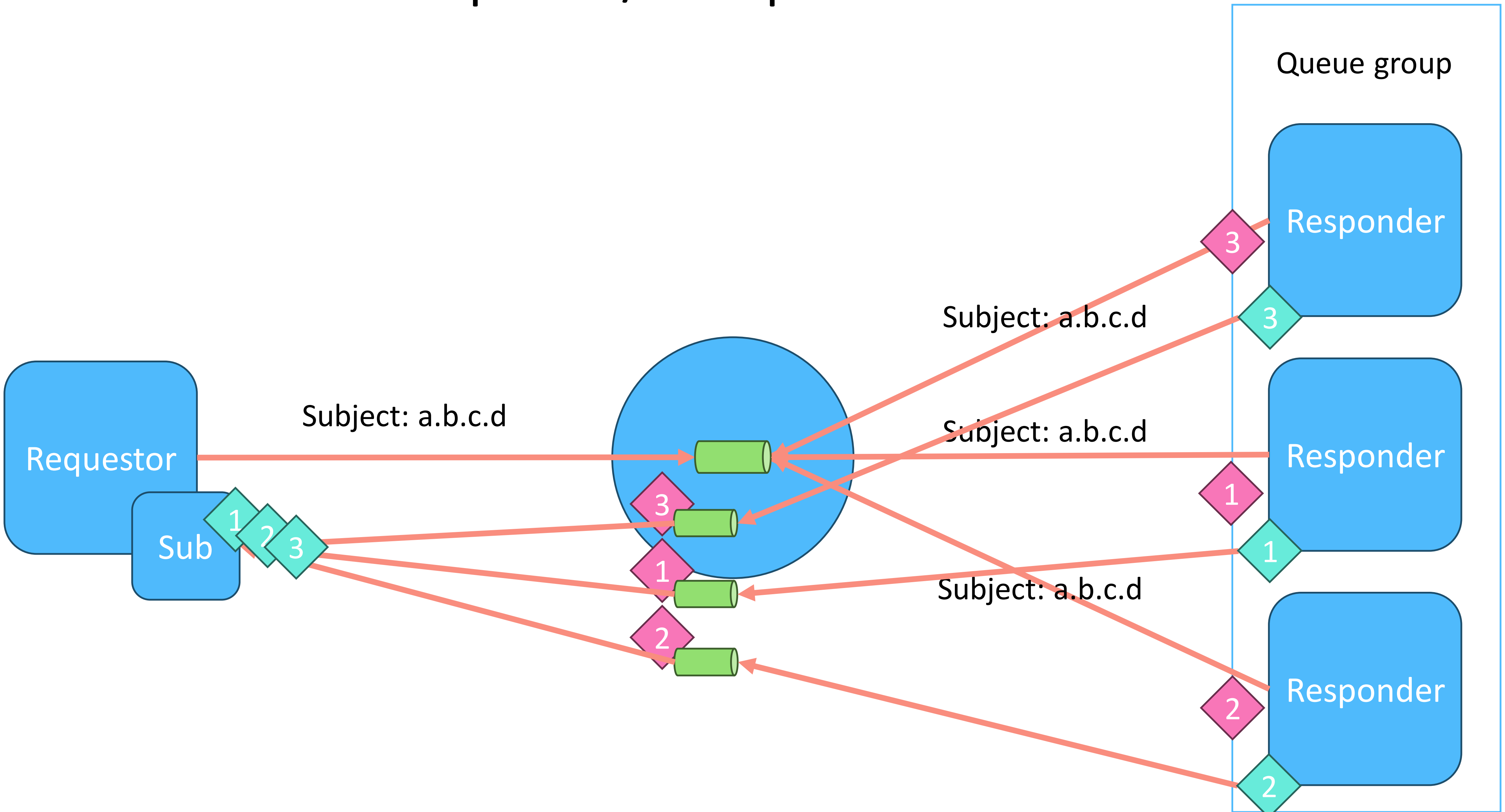
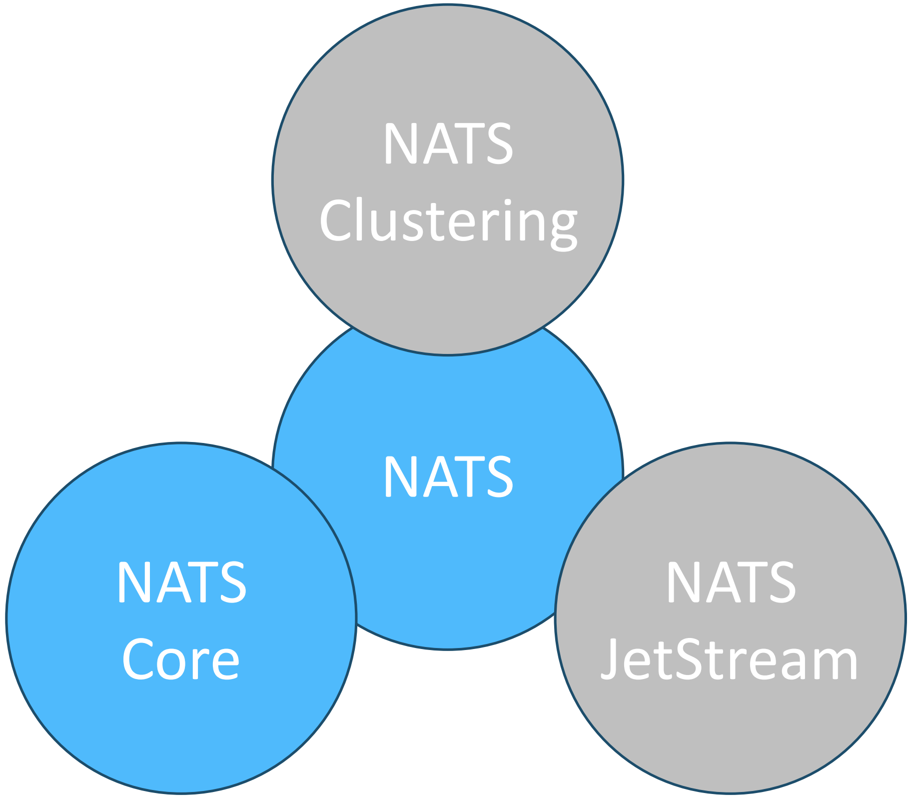
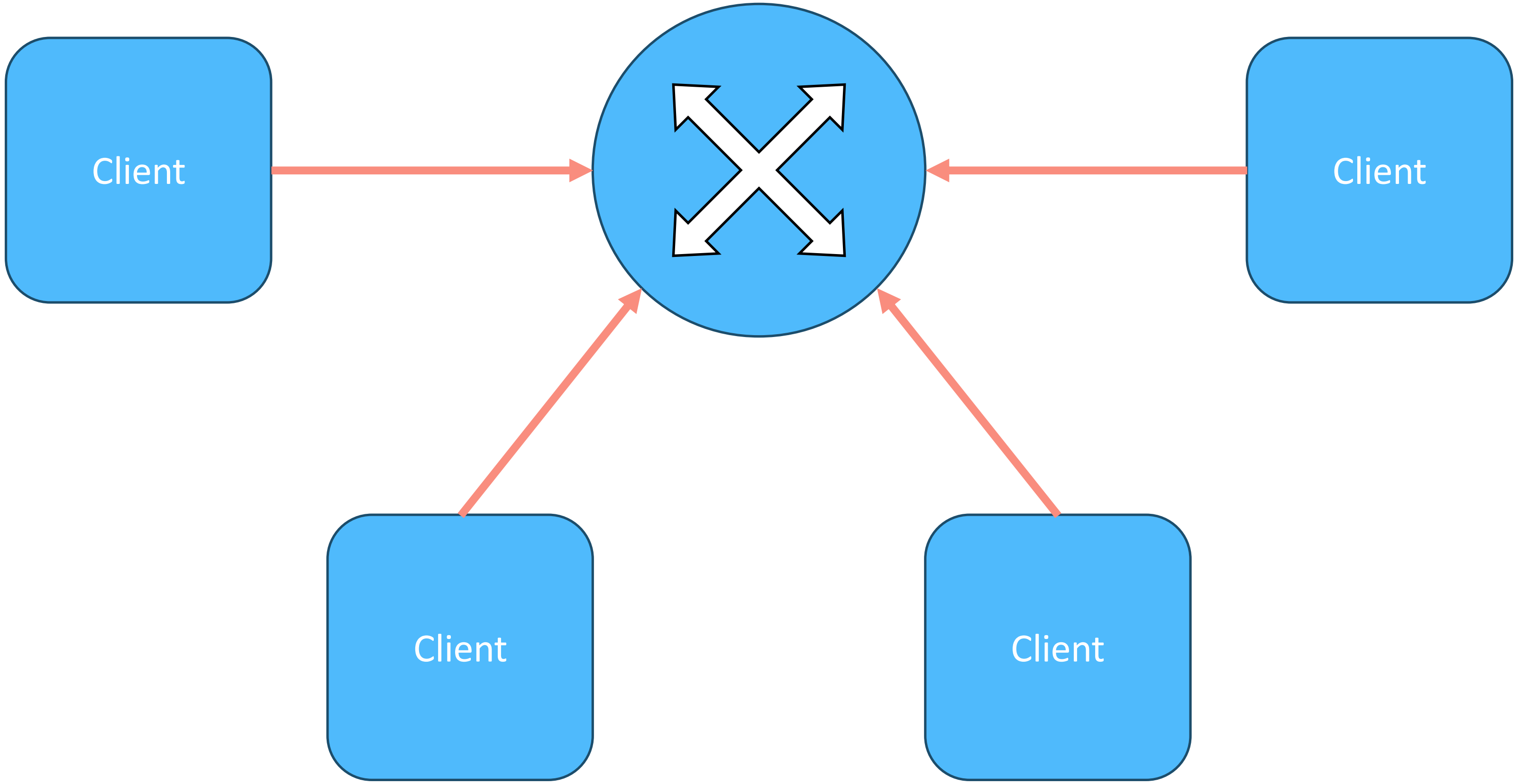


NATS Core: Messaging

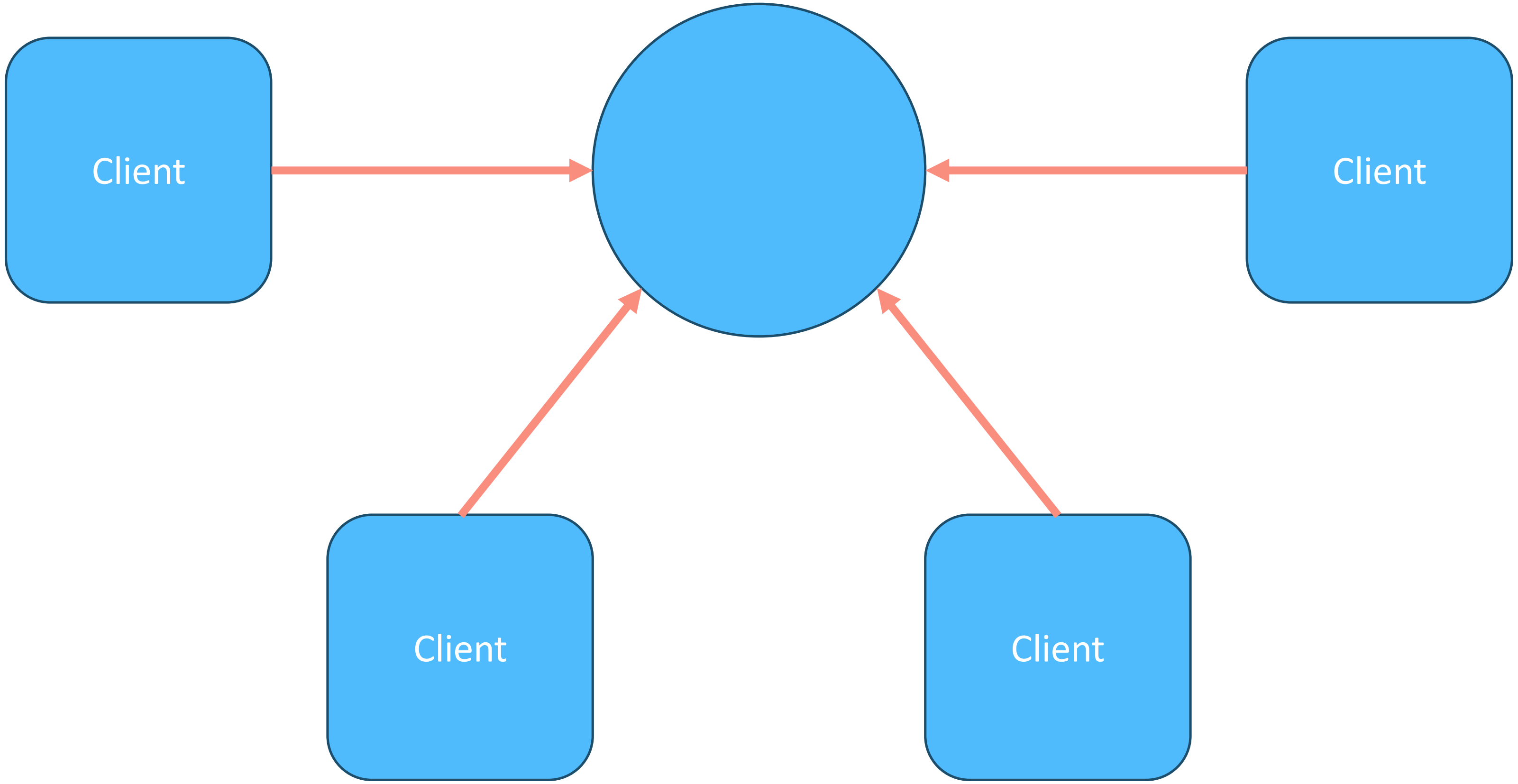


NATS Clustering

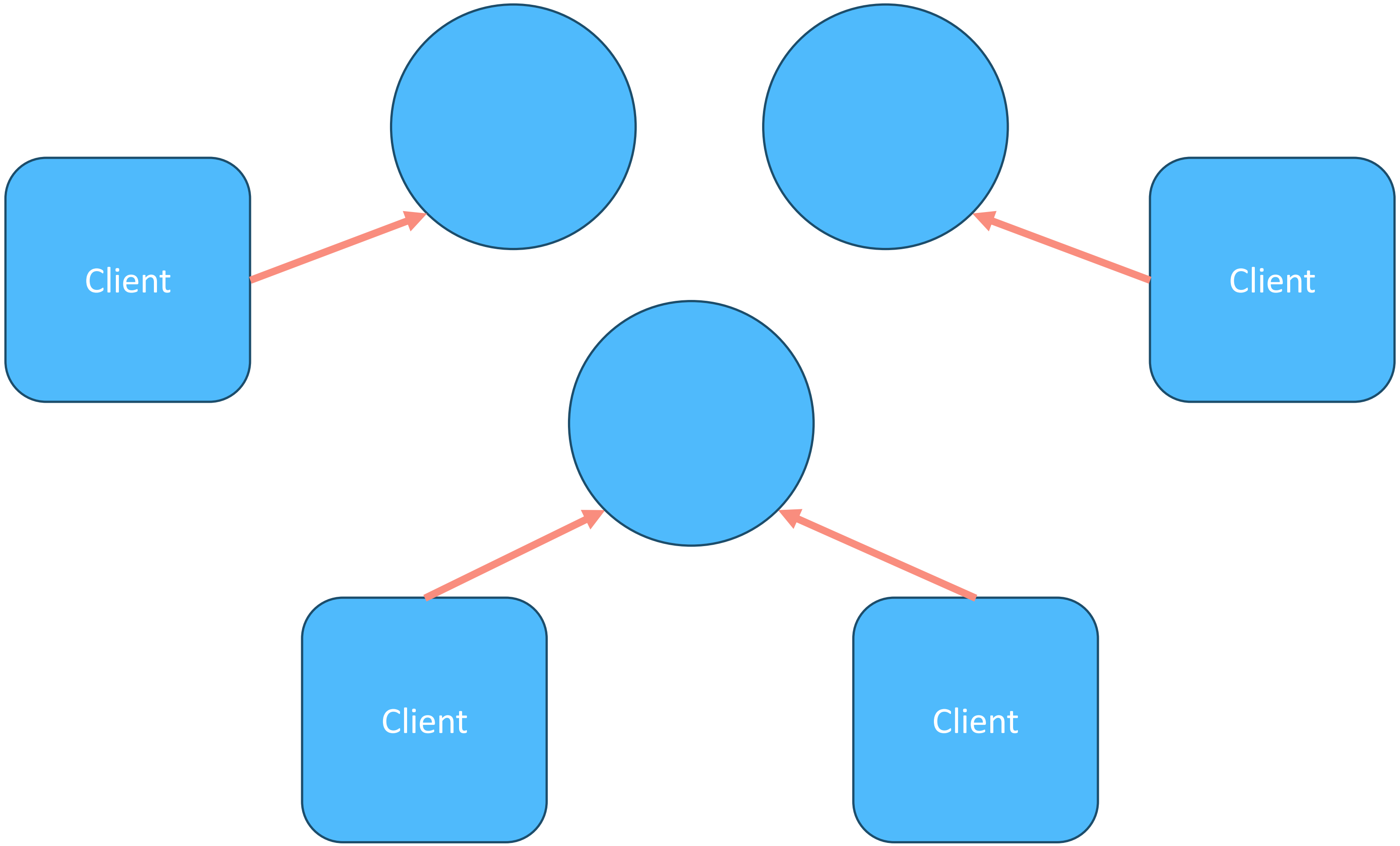
NATS Single



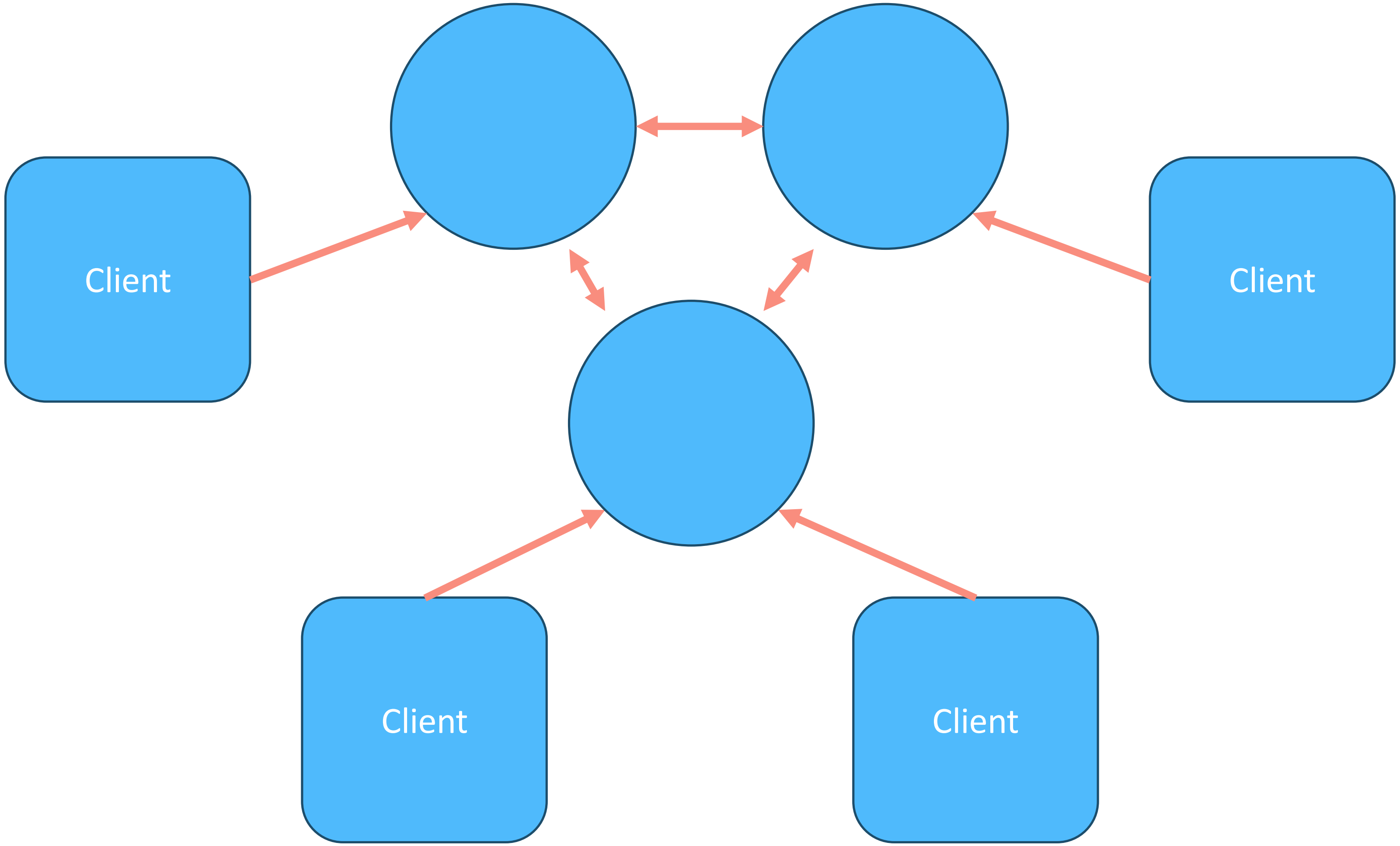
NATS Single



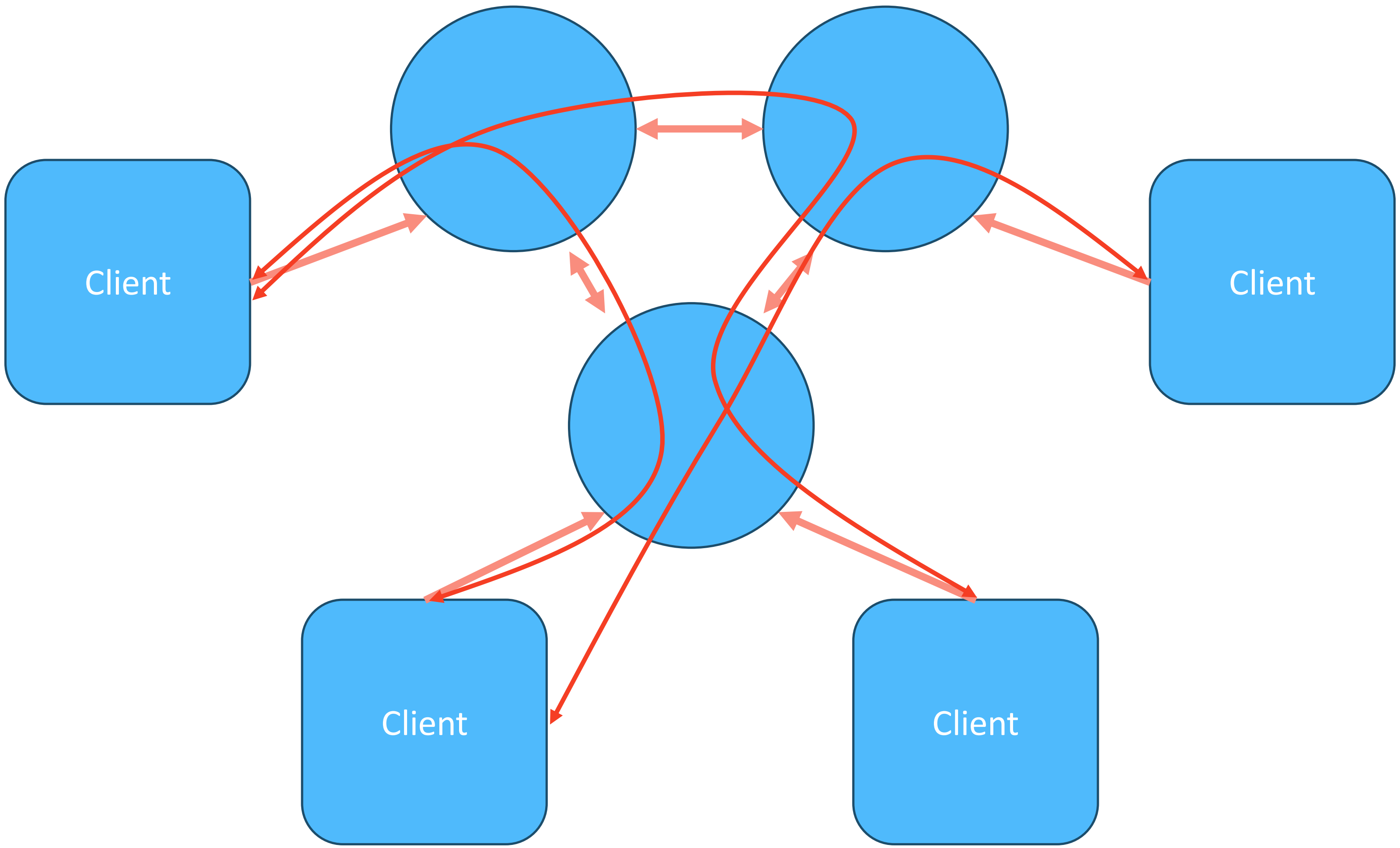
NATS Cluster



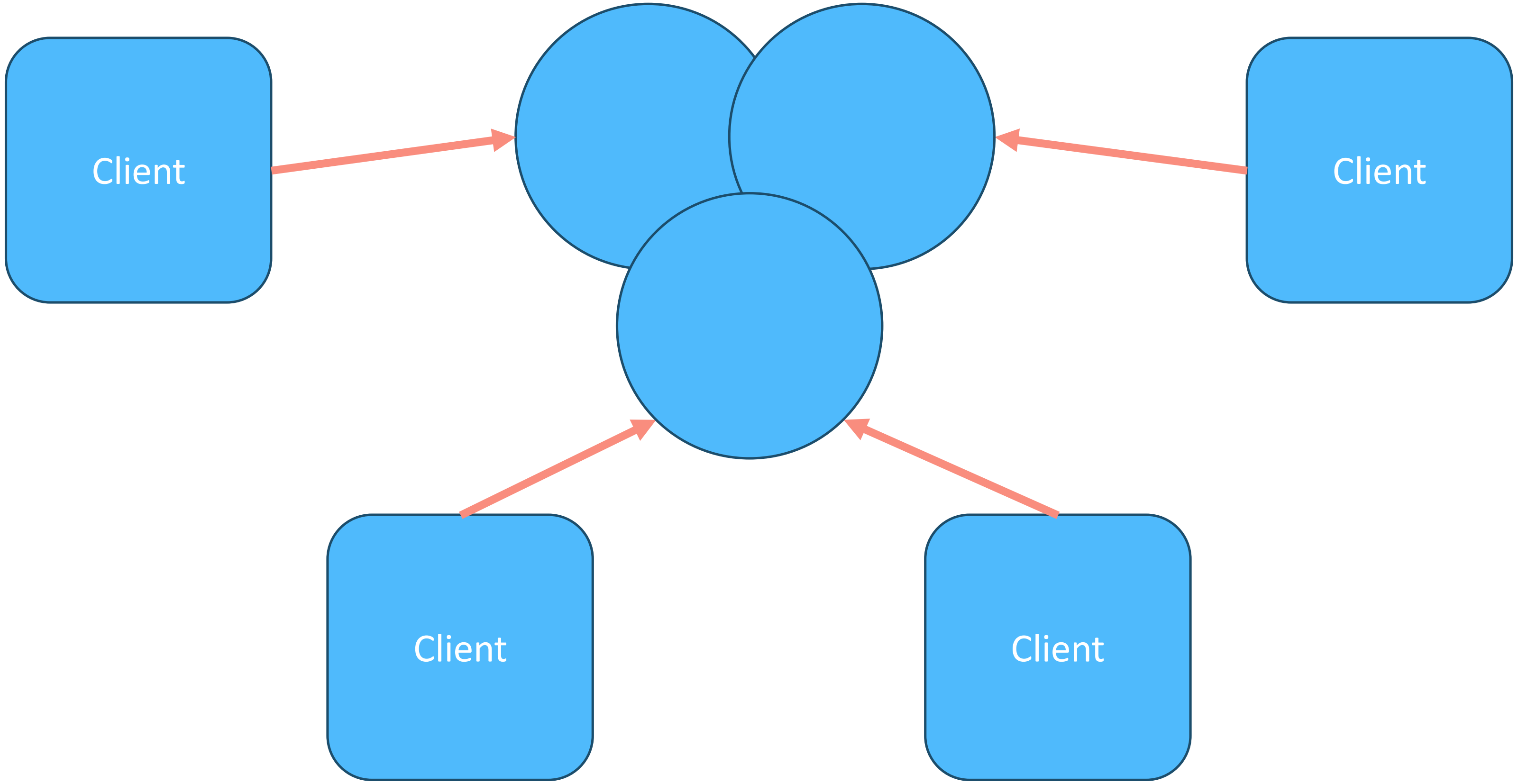
NATS Cluster



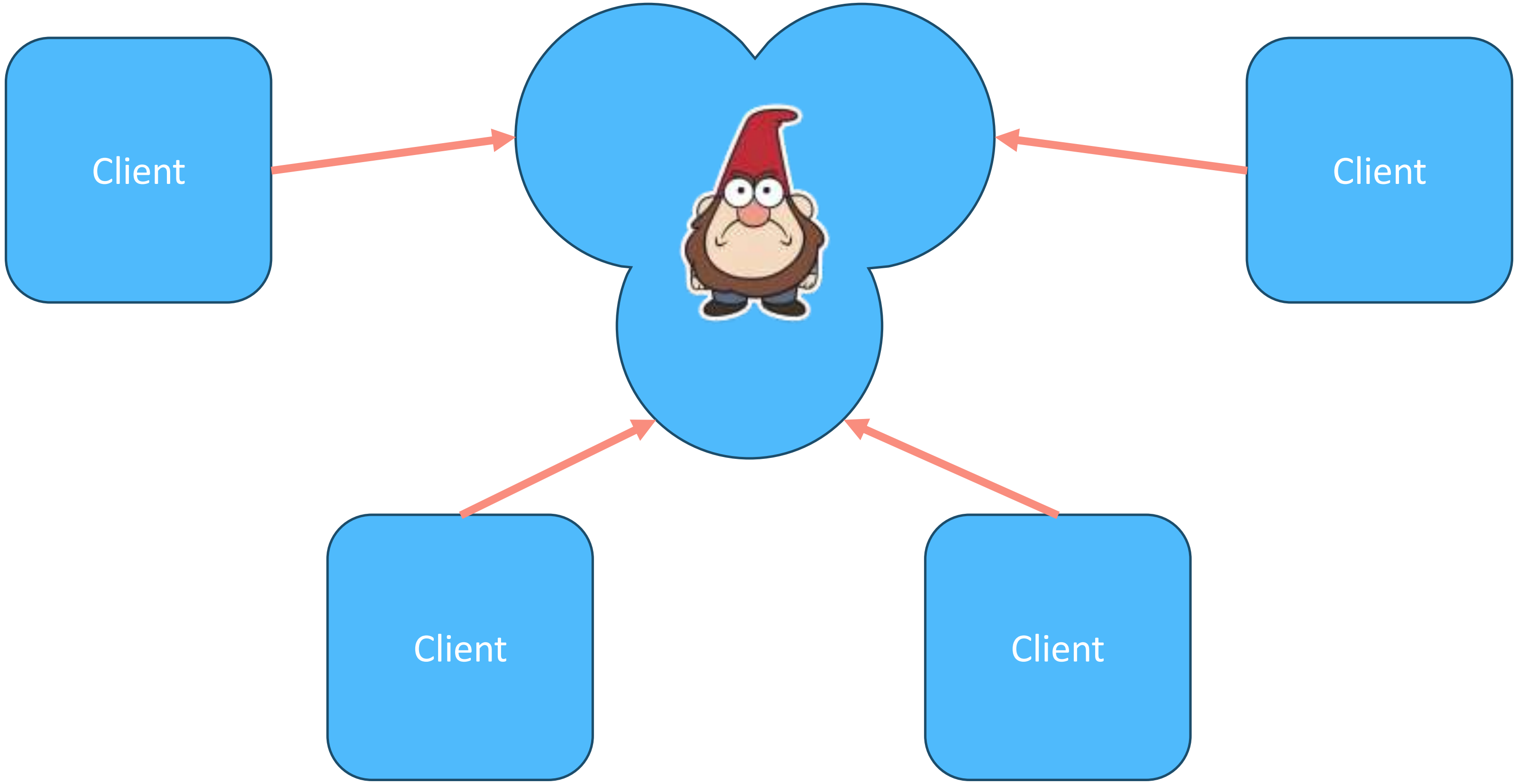
NATS Cluster Routing



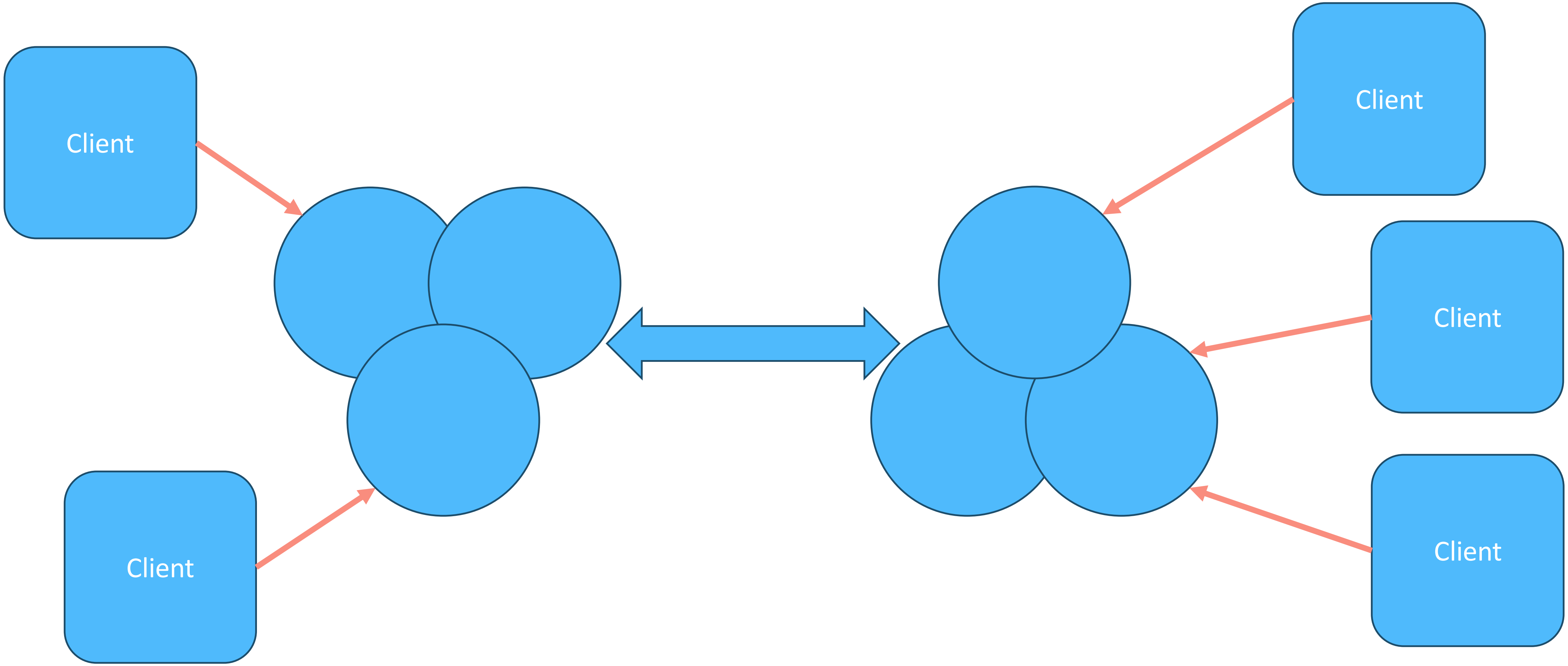
NATS Cluster



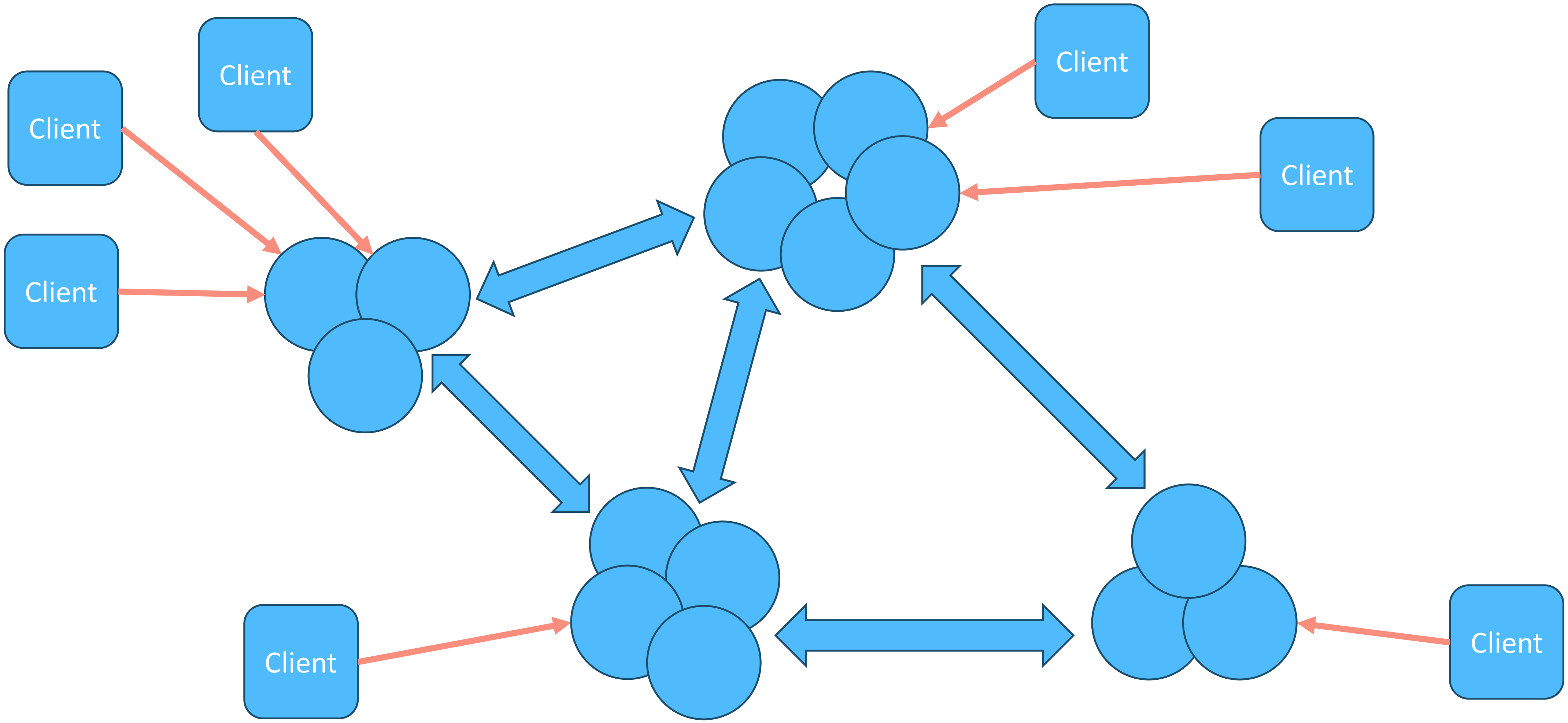
NATS Cluster



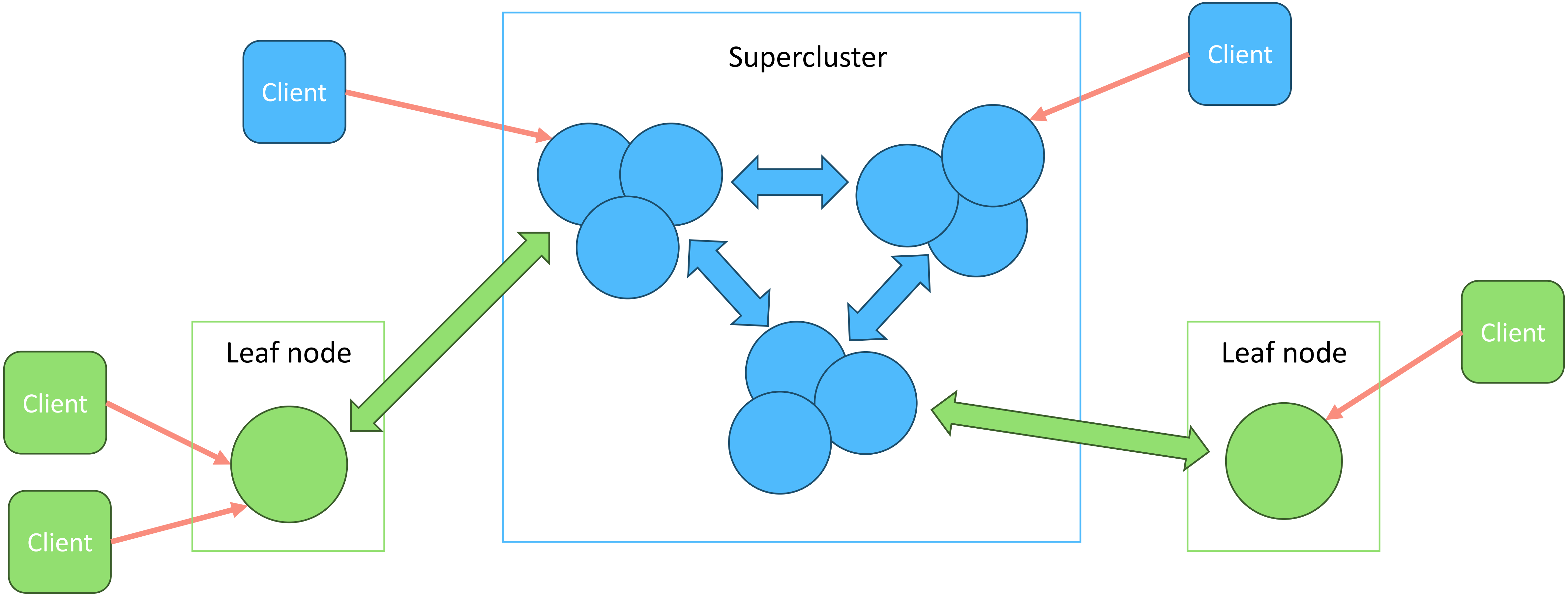
NATS Supercluster (Cluster of clusters)



NATS Supercluster



NATS Supercluster and Leaf Nodes



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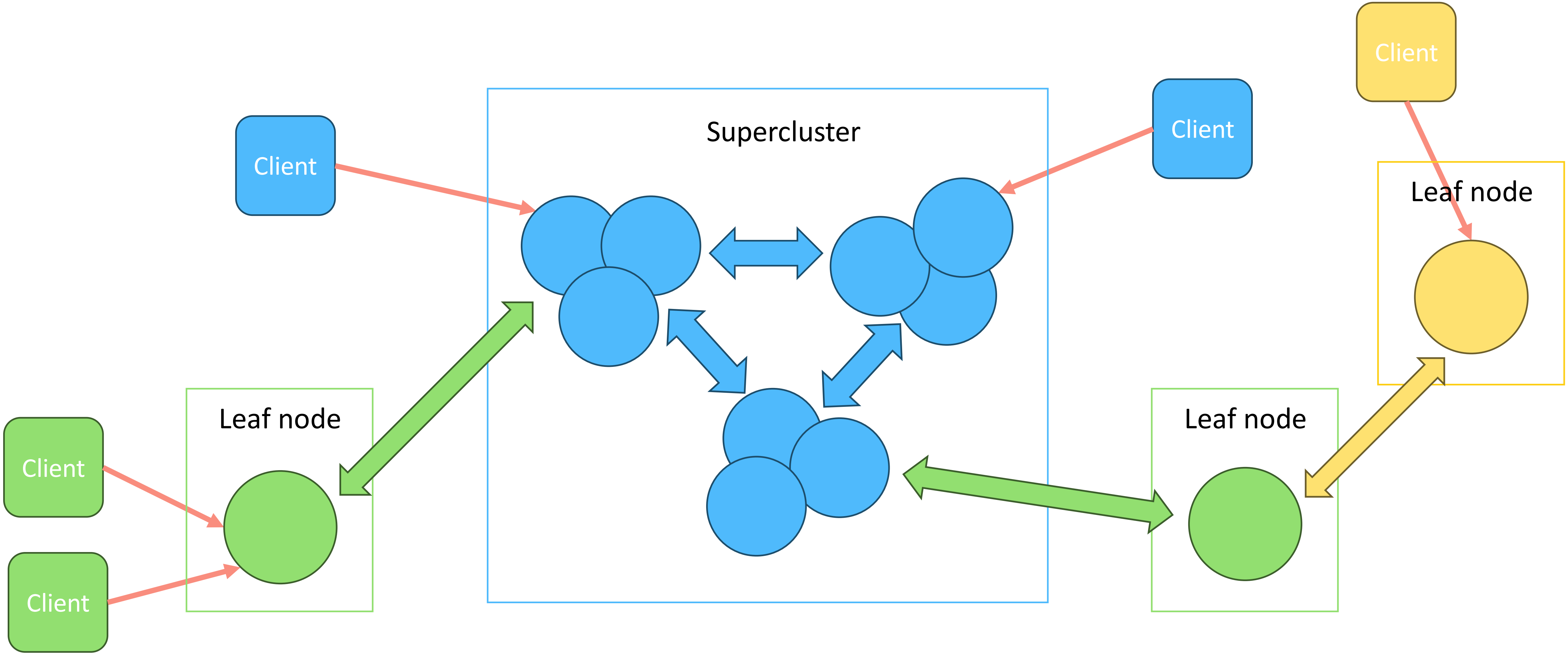
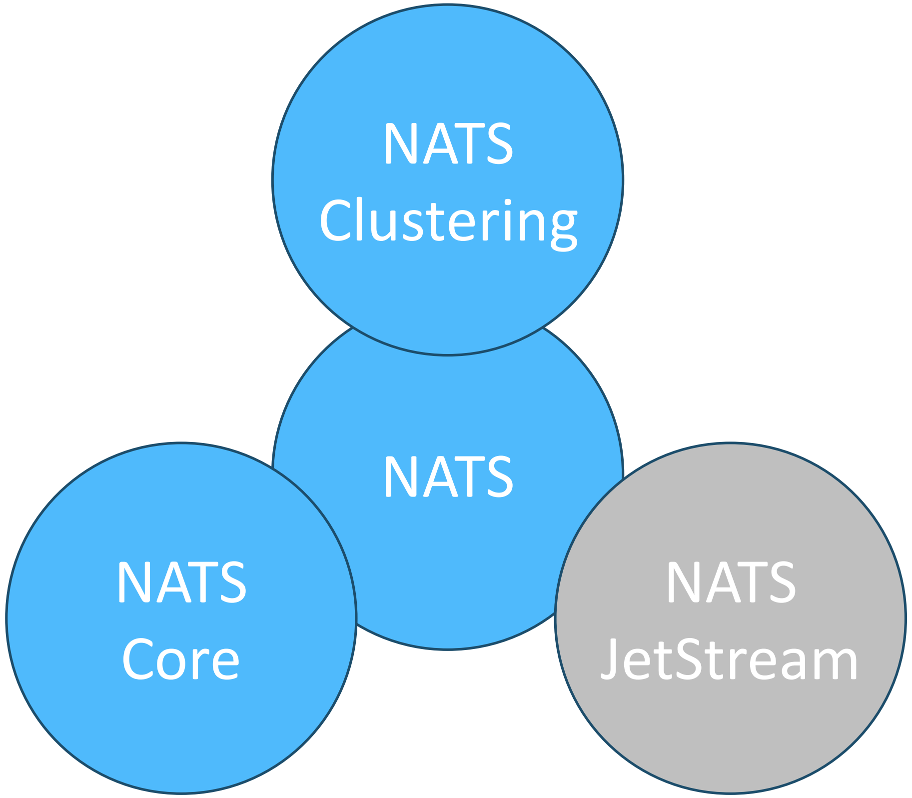


NATS Clustering: Topology



NATS JetStream

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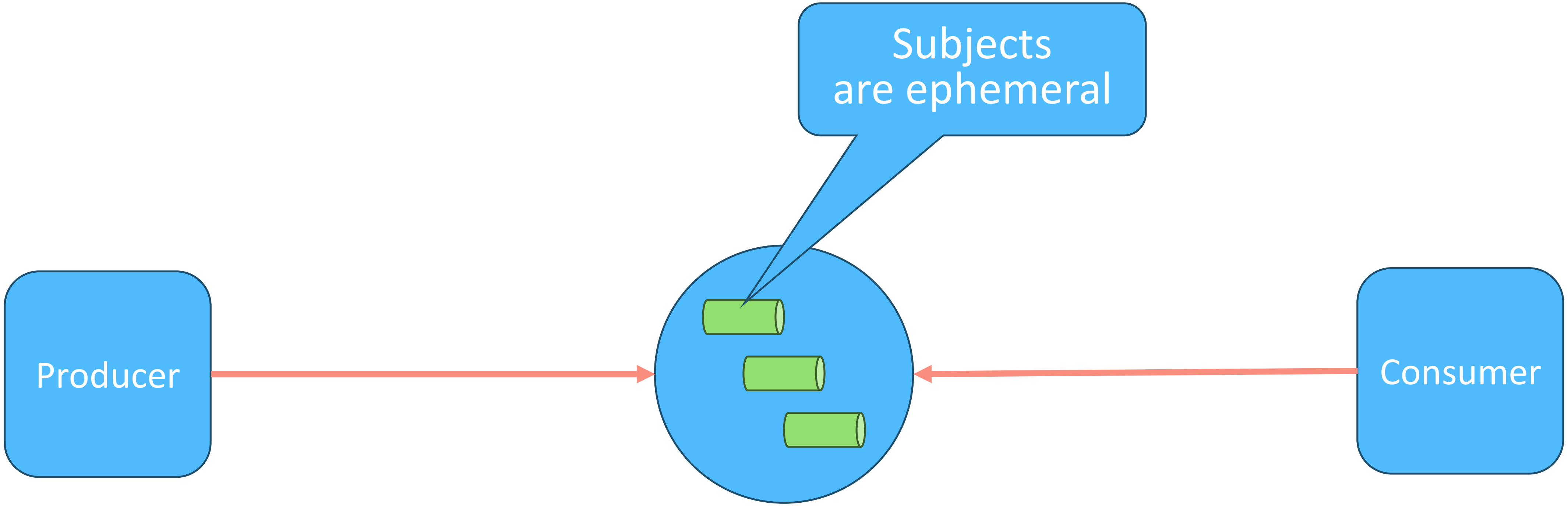
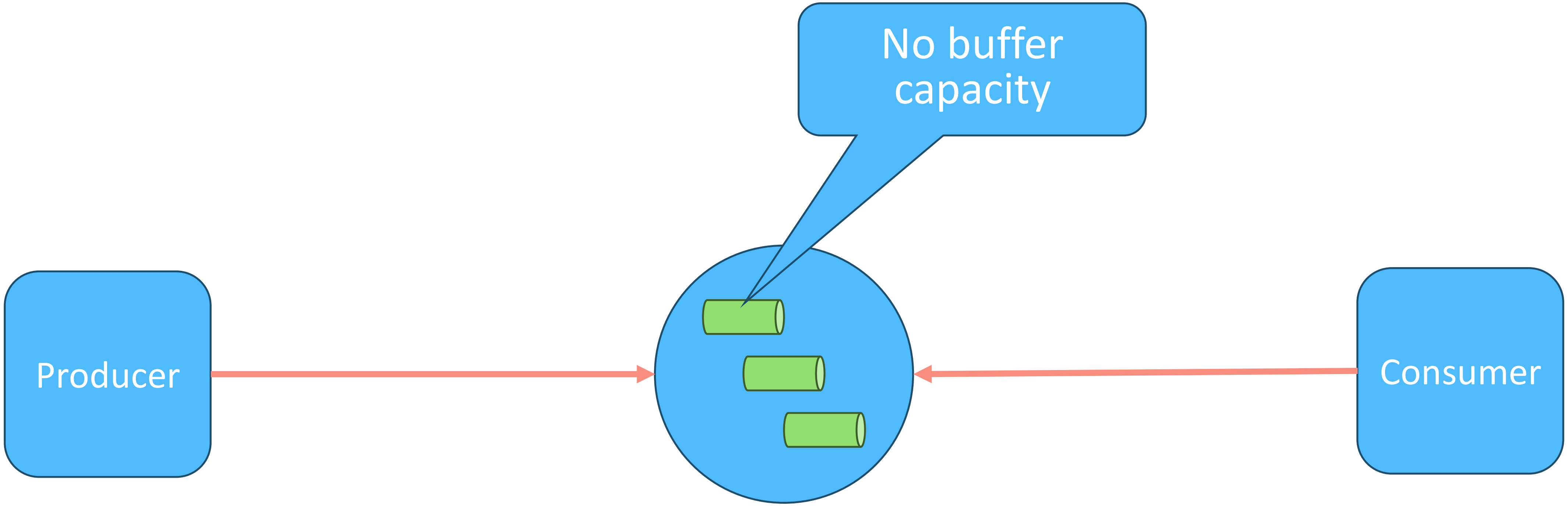
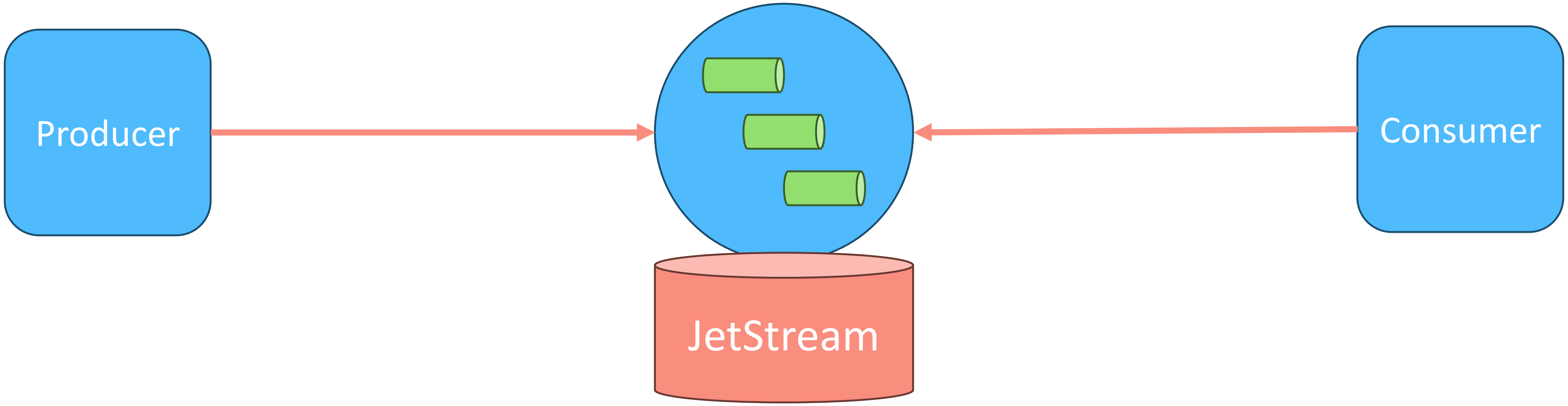


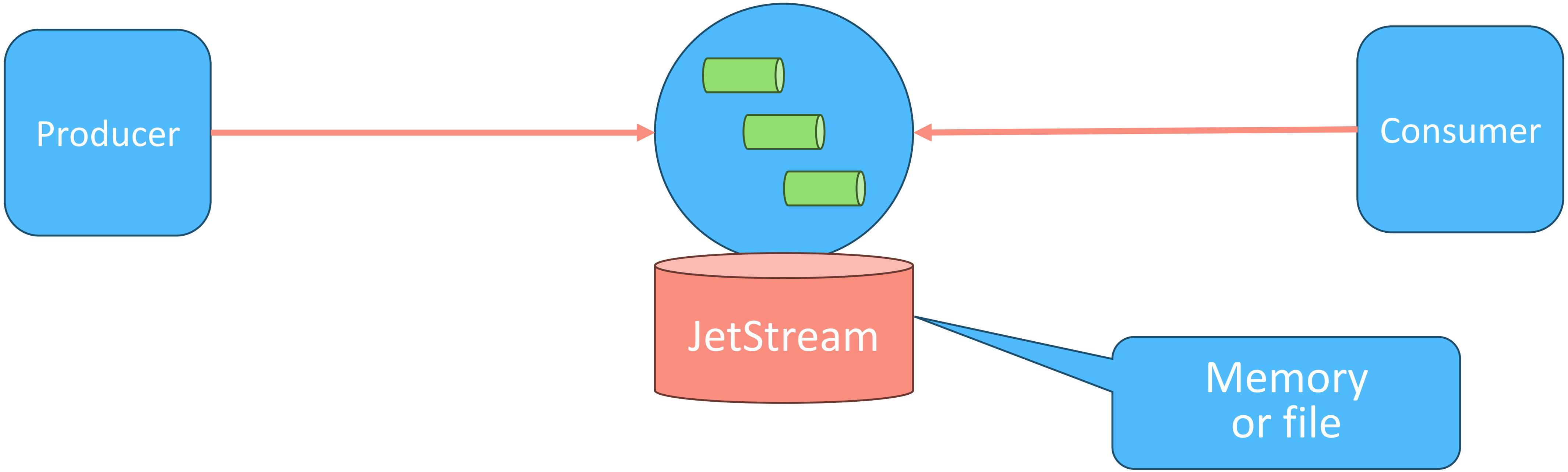
NATS JetStream



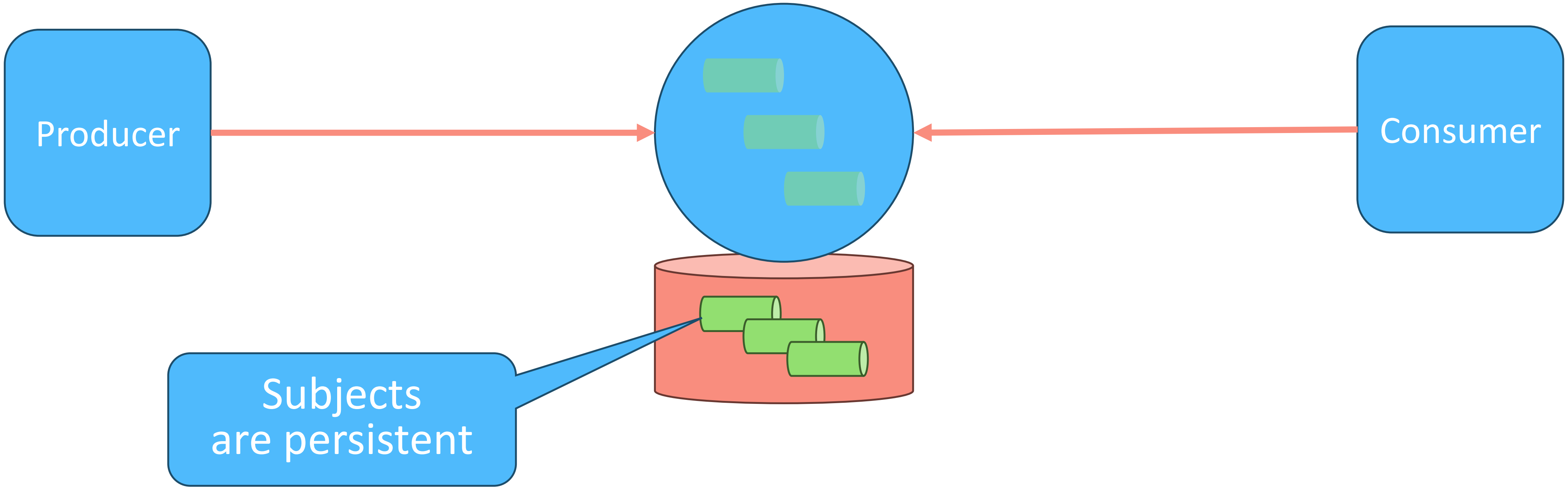
NATS JetStream



NATS JetStream



NATS JetStream



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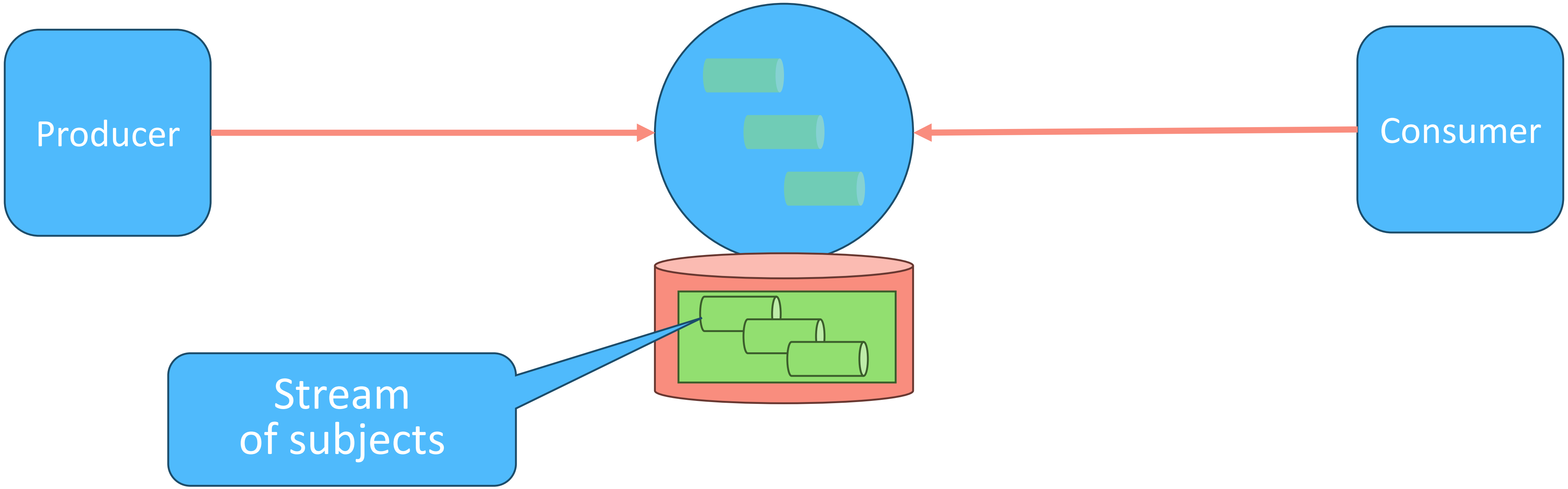

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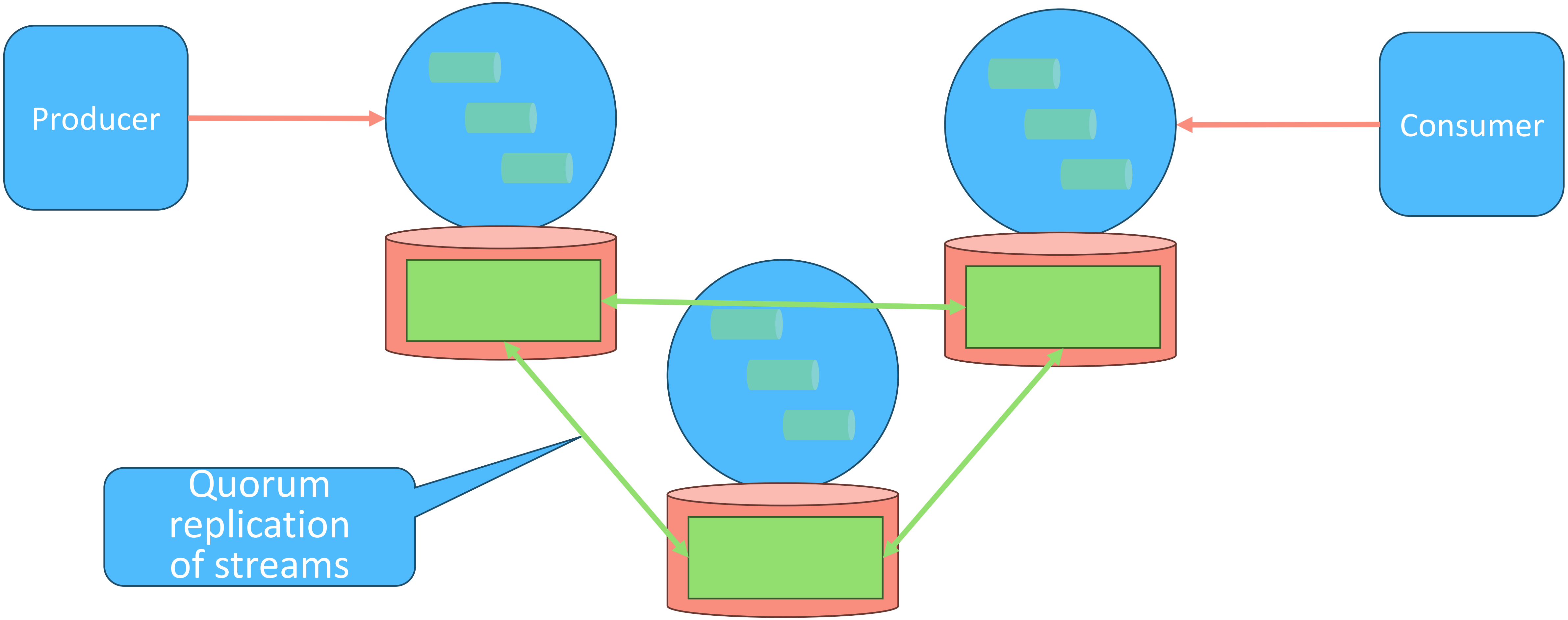
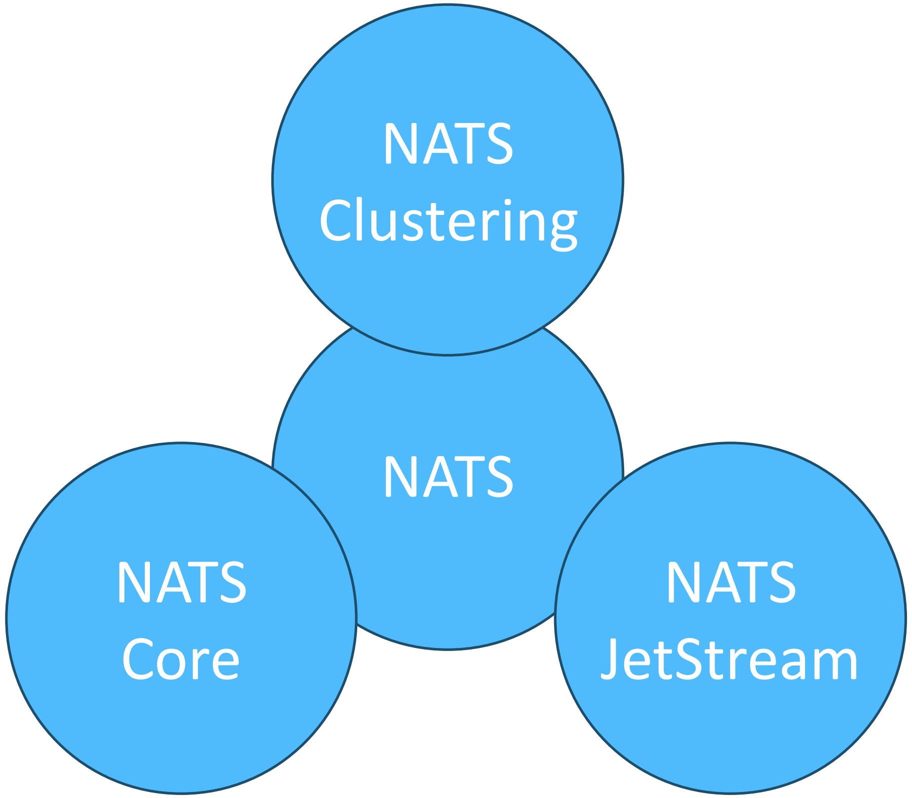
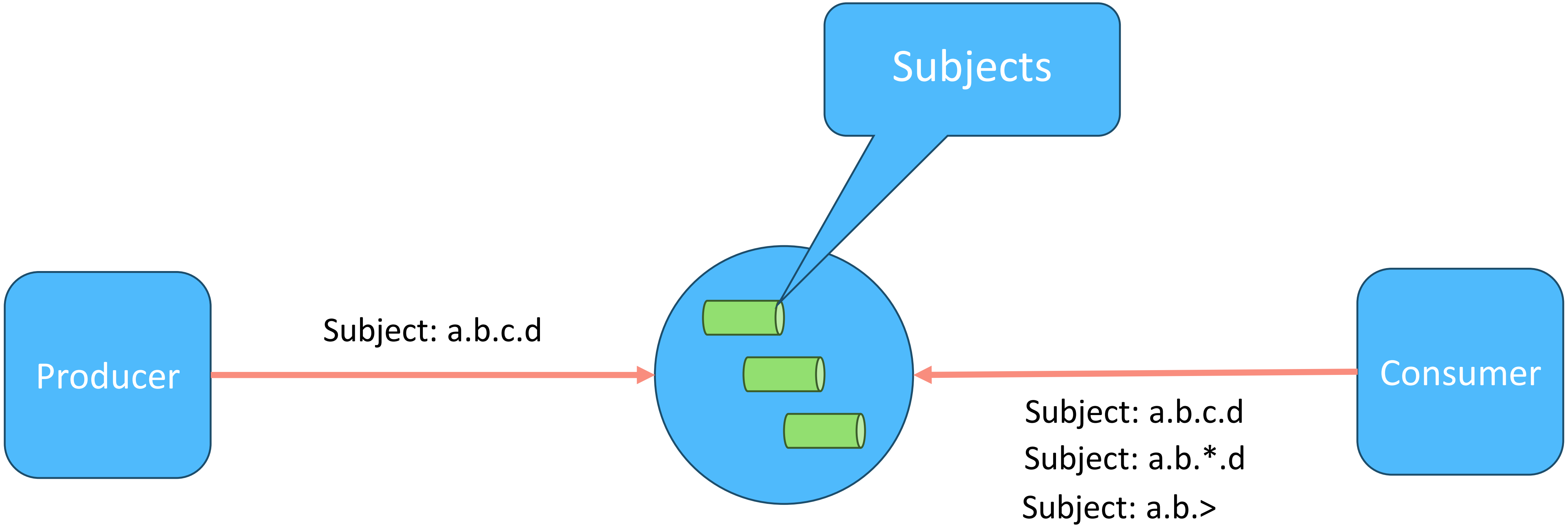


NATS JetStream: Persistence & Streaming

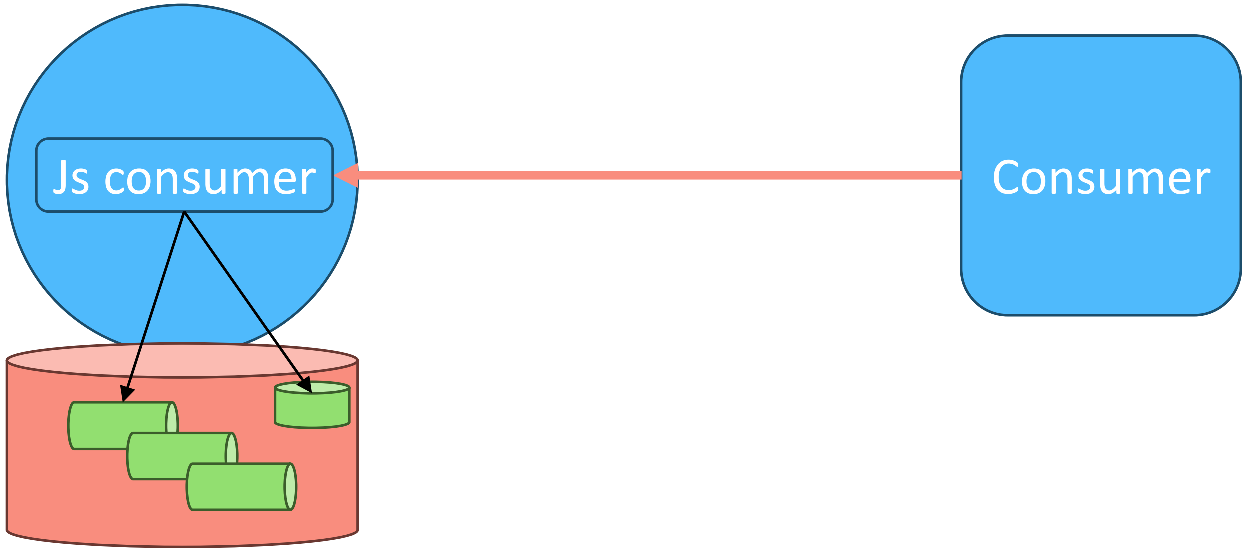


JetStream Consumers

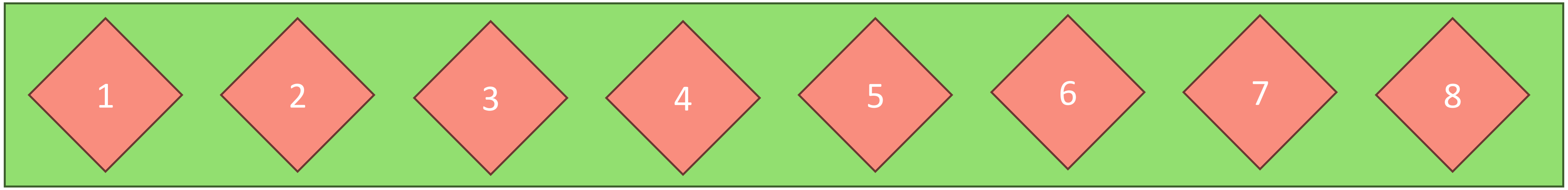
NATS Consumer



NATS JetStream Consumer (Durable)

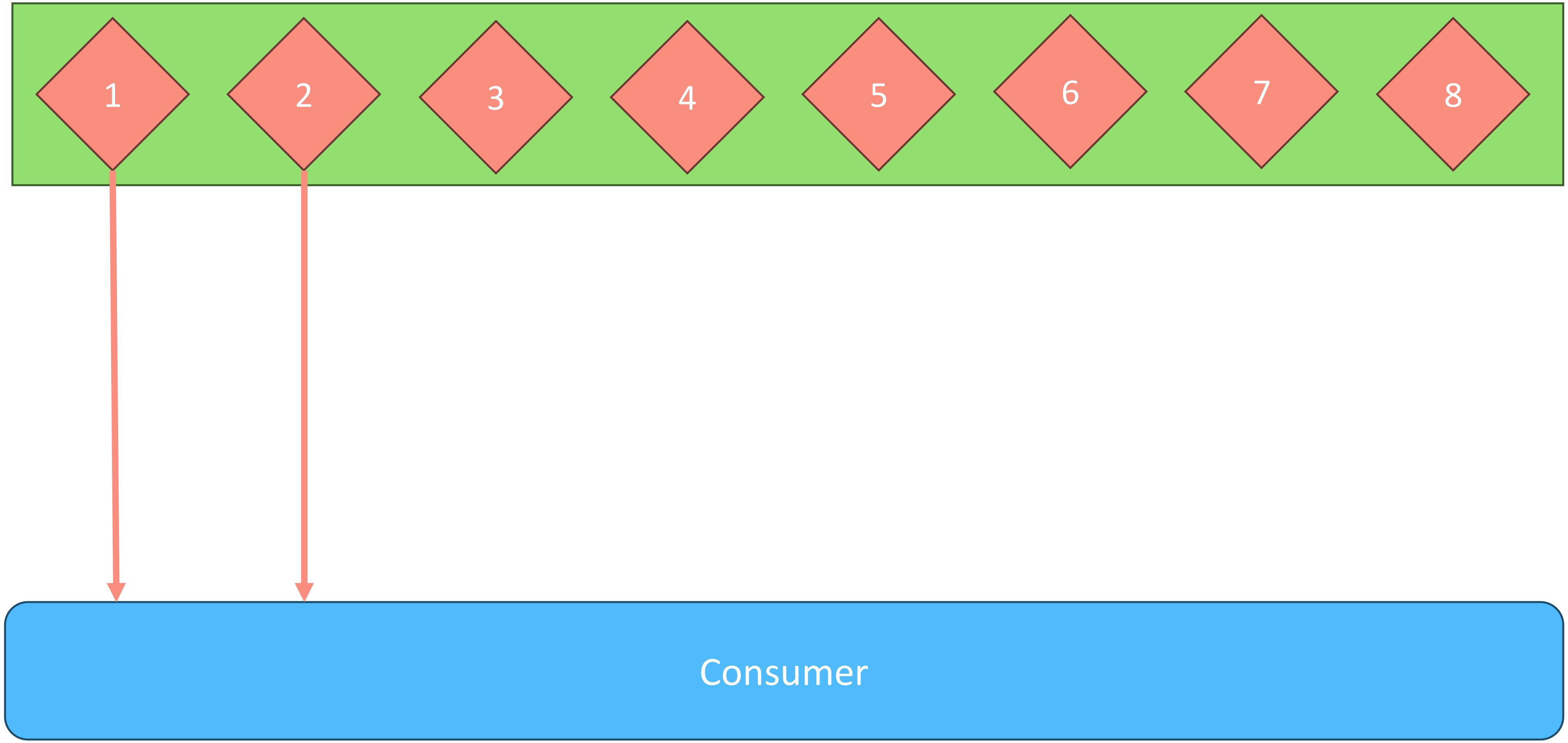


Consumer Acks

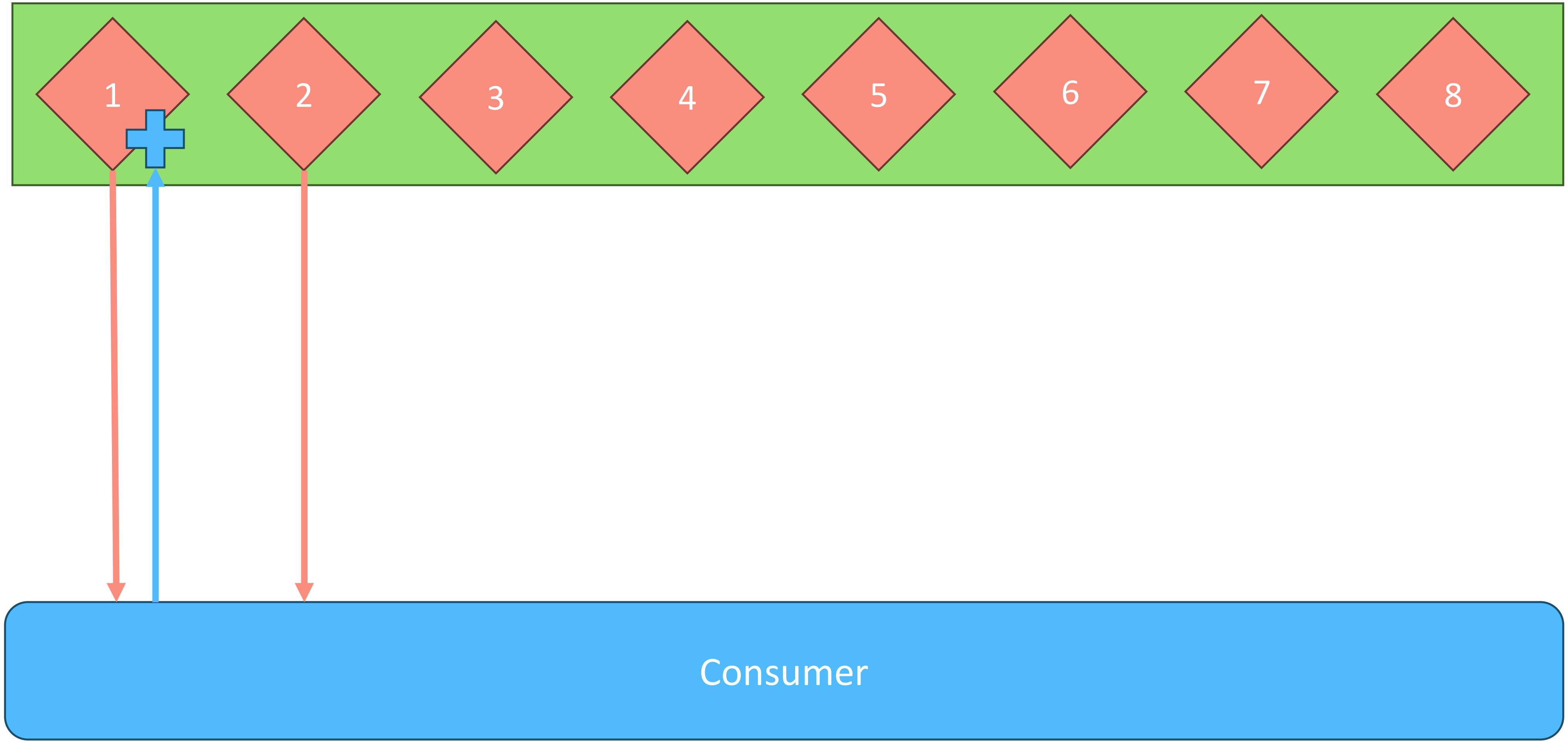


Explicit

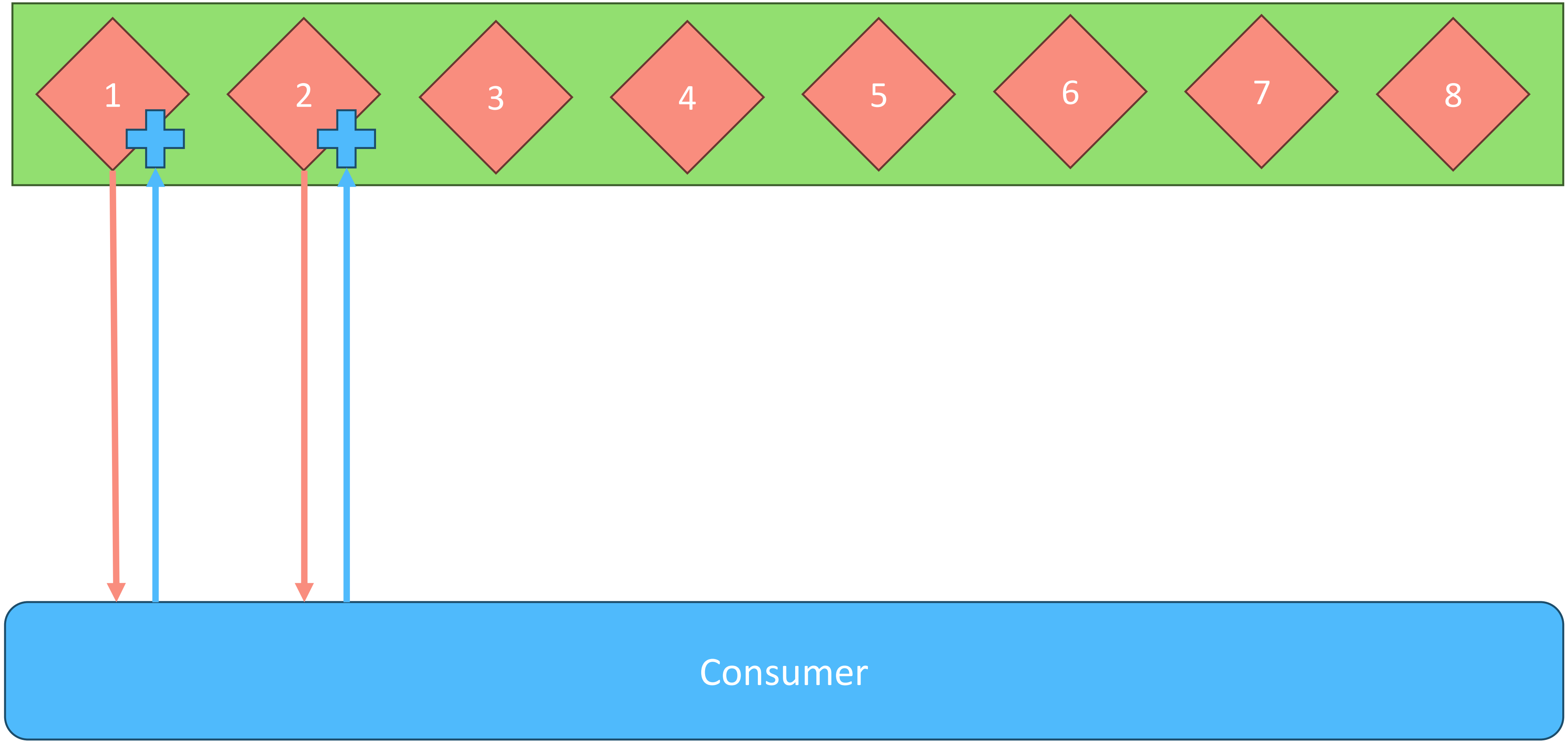
Consumer Acks: Explicit



Consumer Acks: Explicit



Consumer Acks: Explicit



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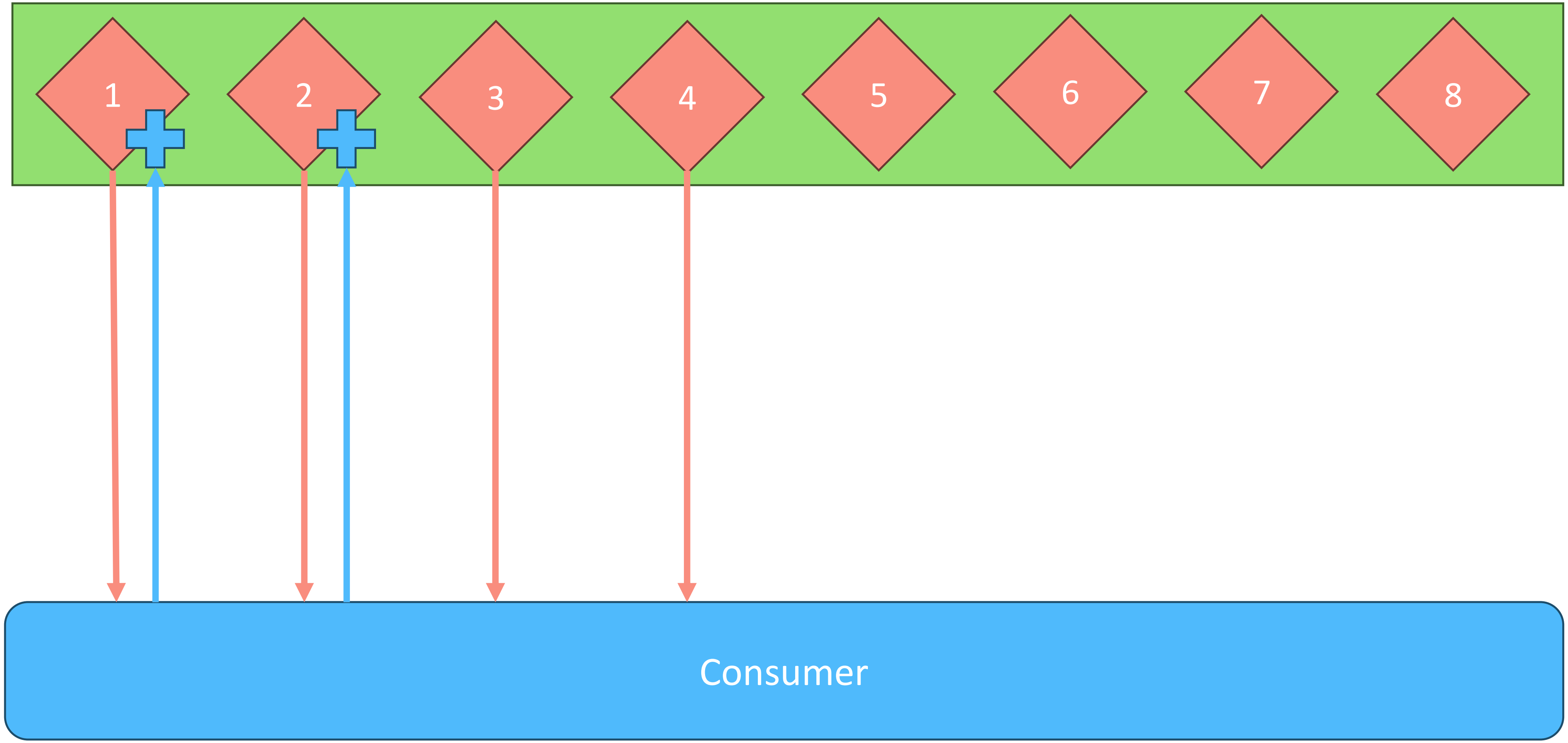

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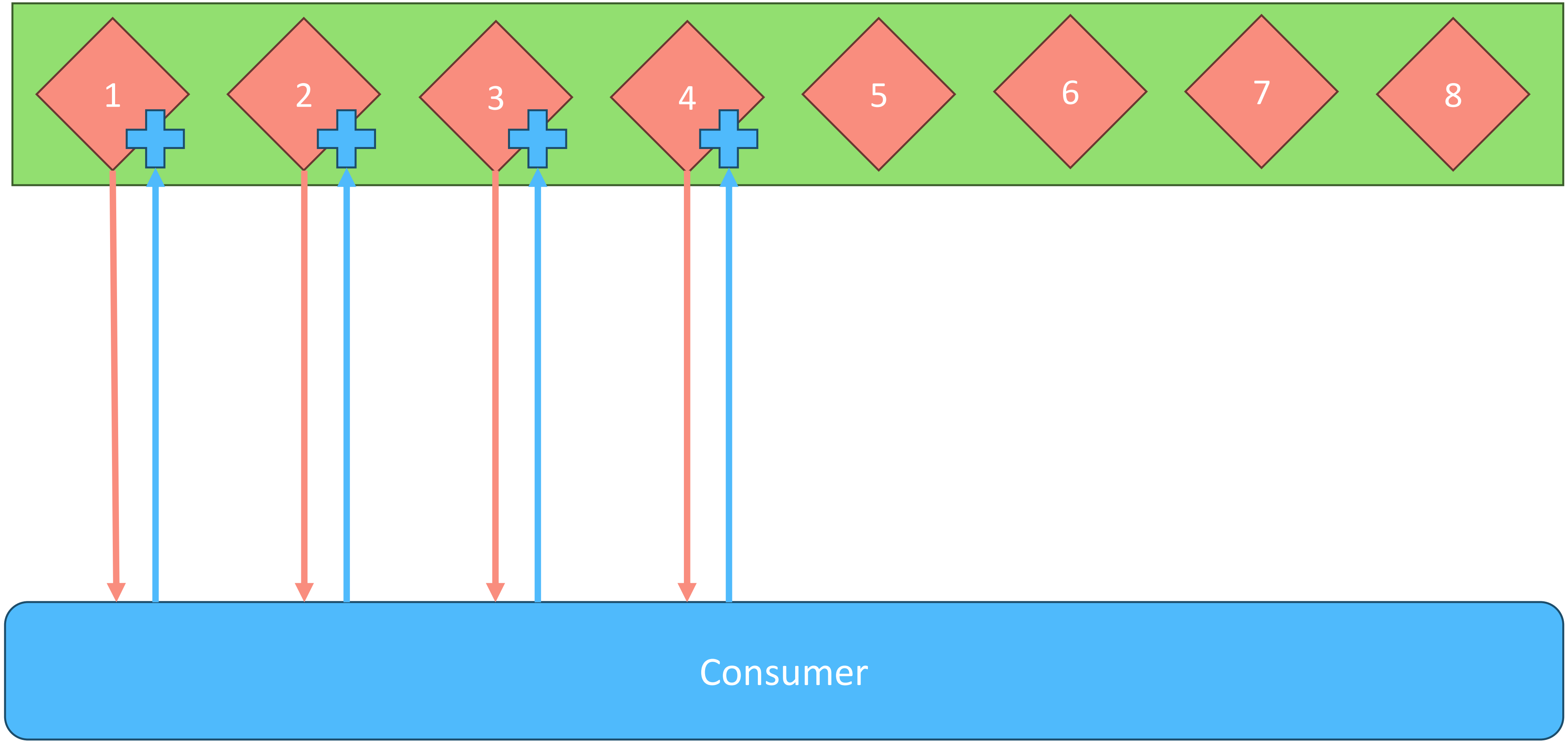

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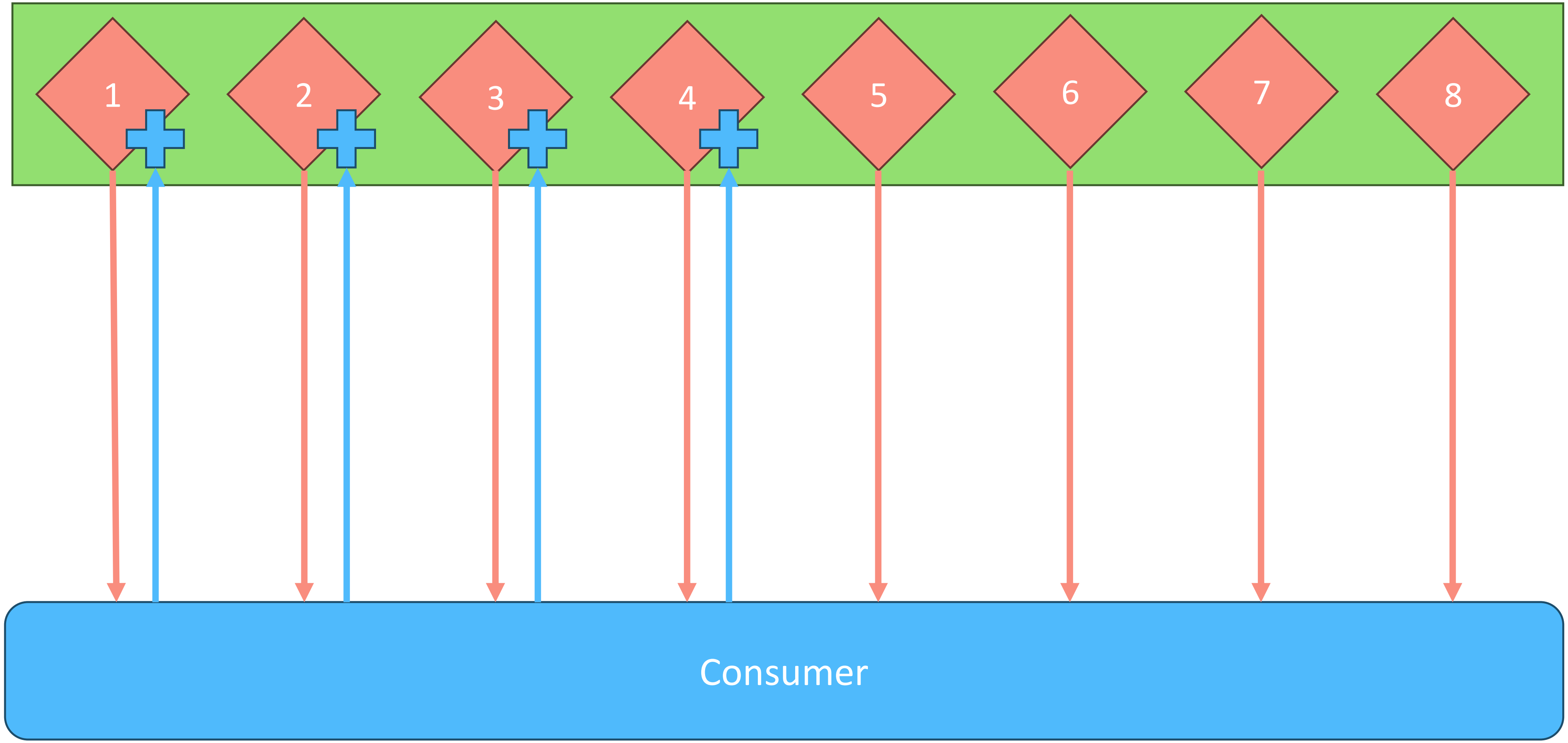

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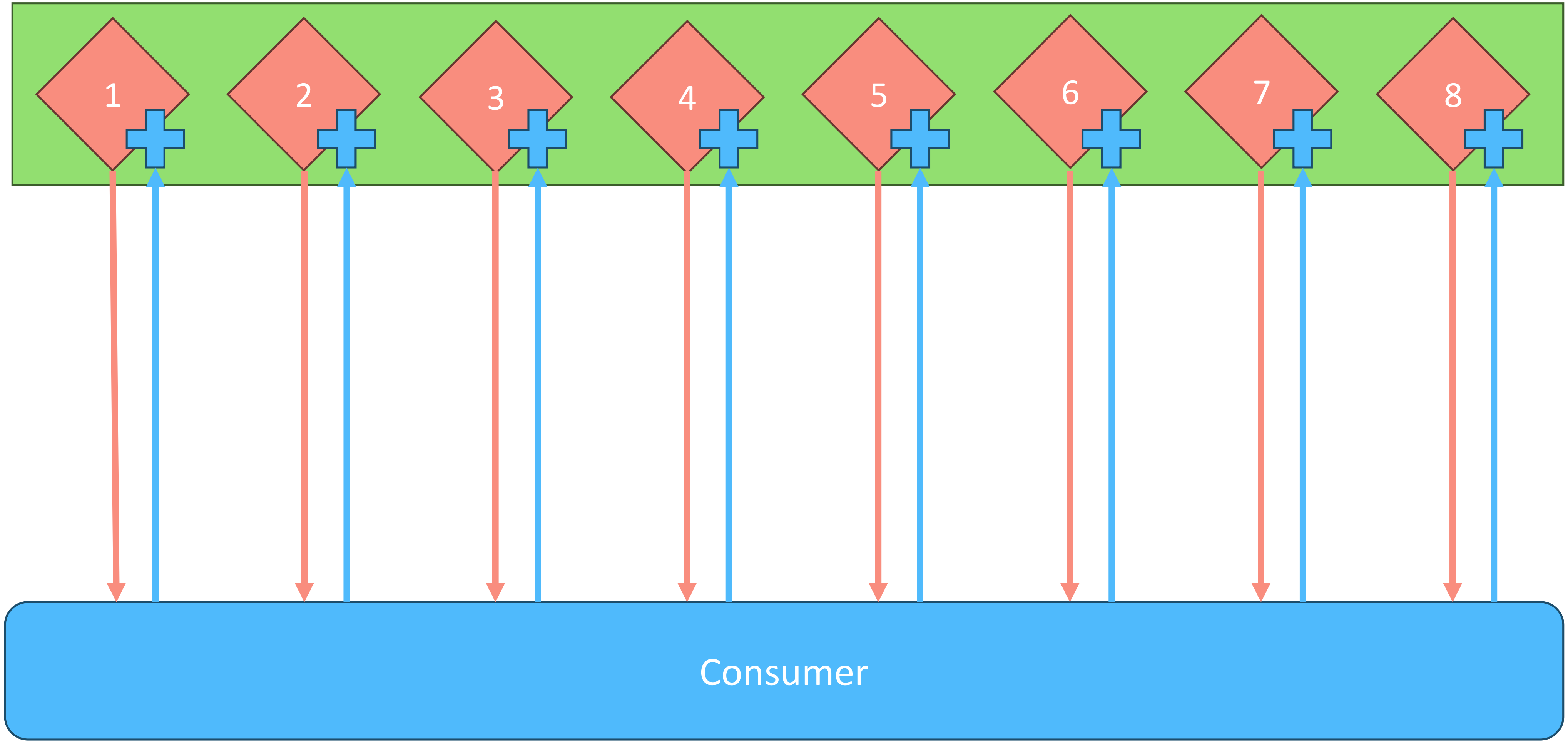

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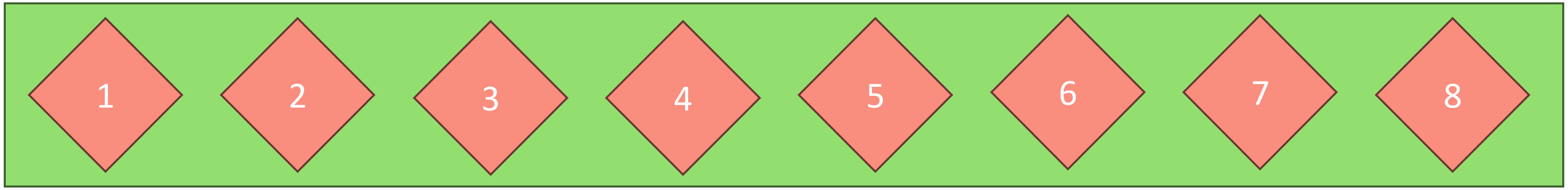
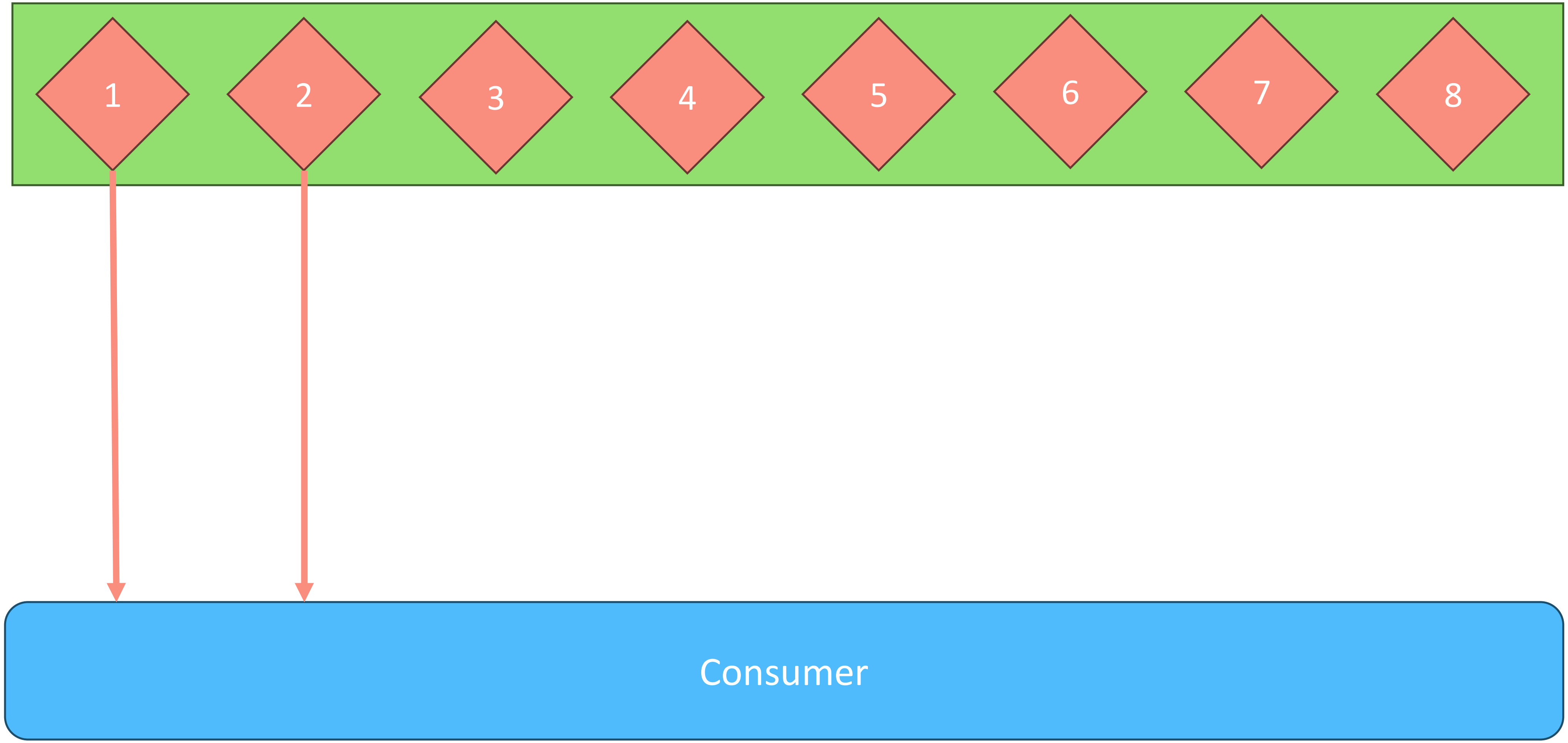


All

Consumer Acks: All



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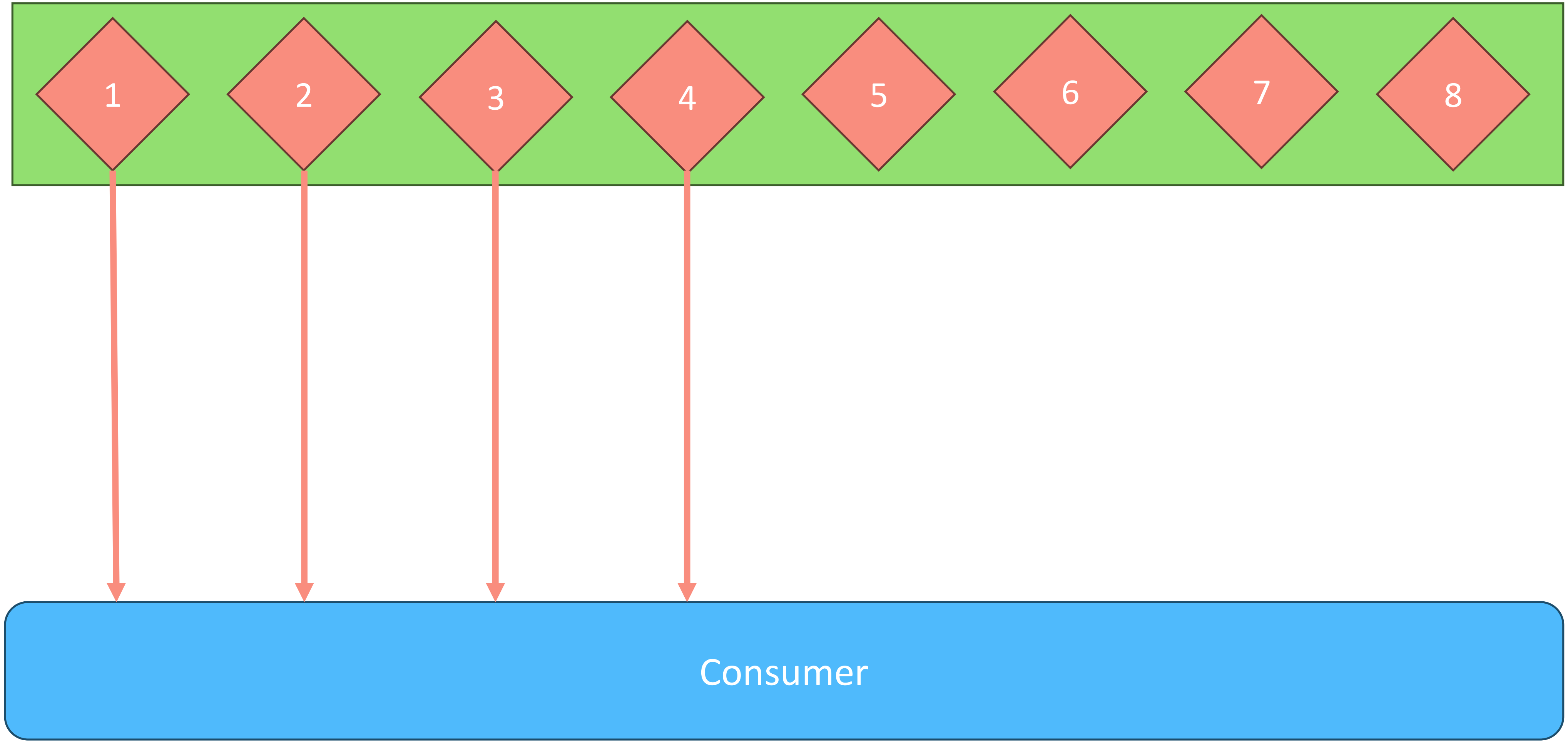

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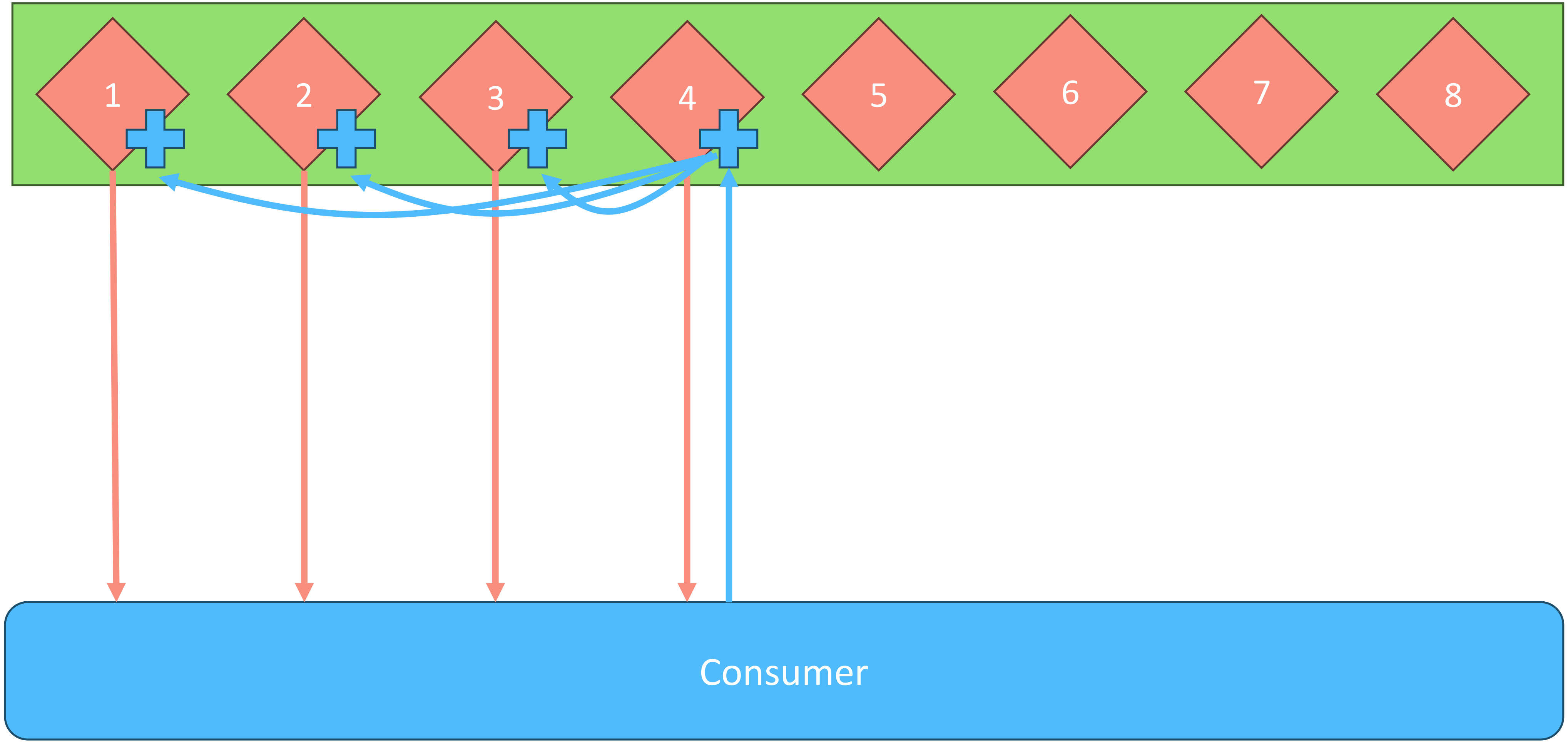

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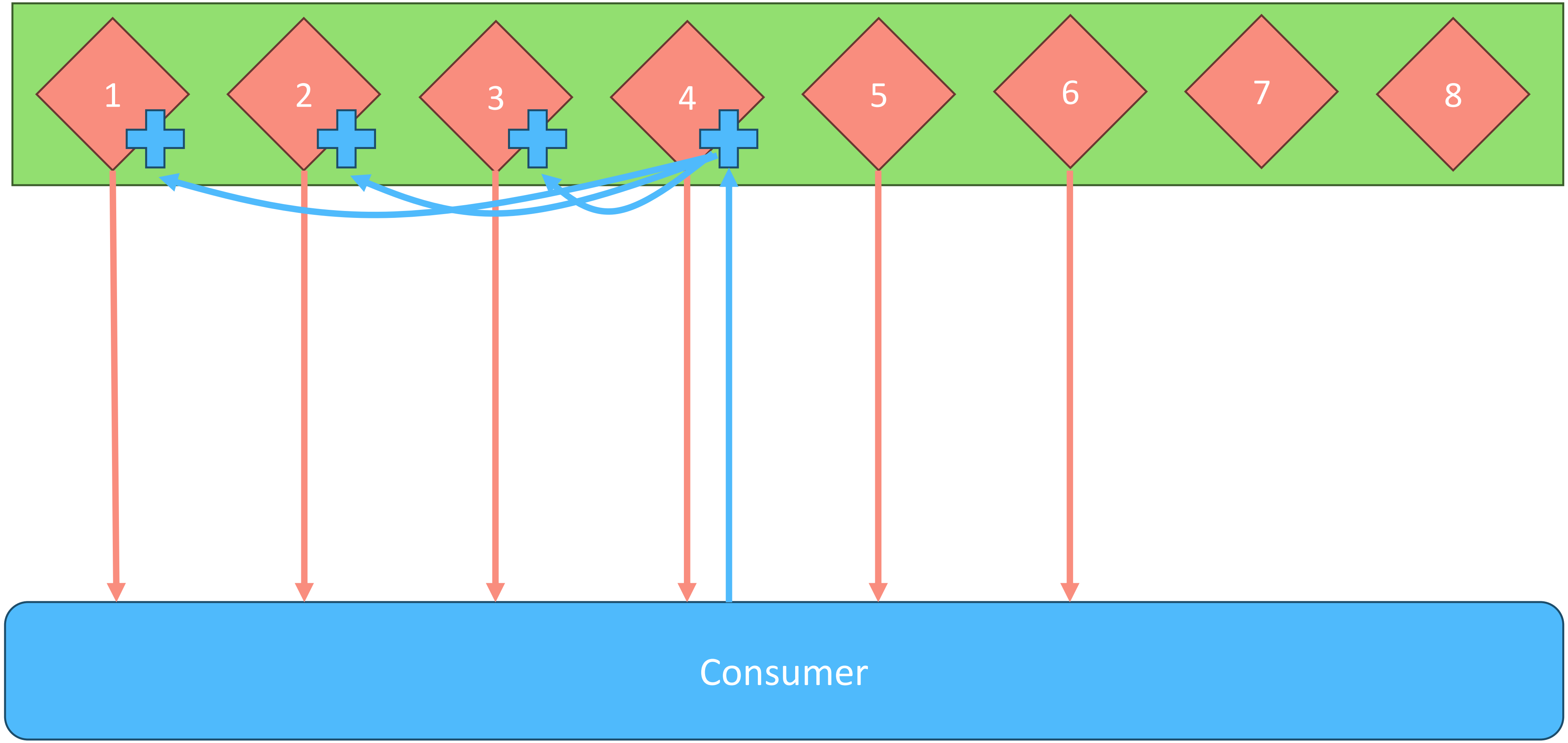

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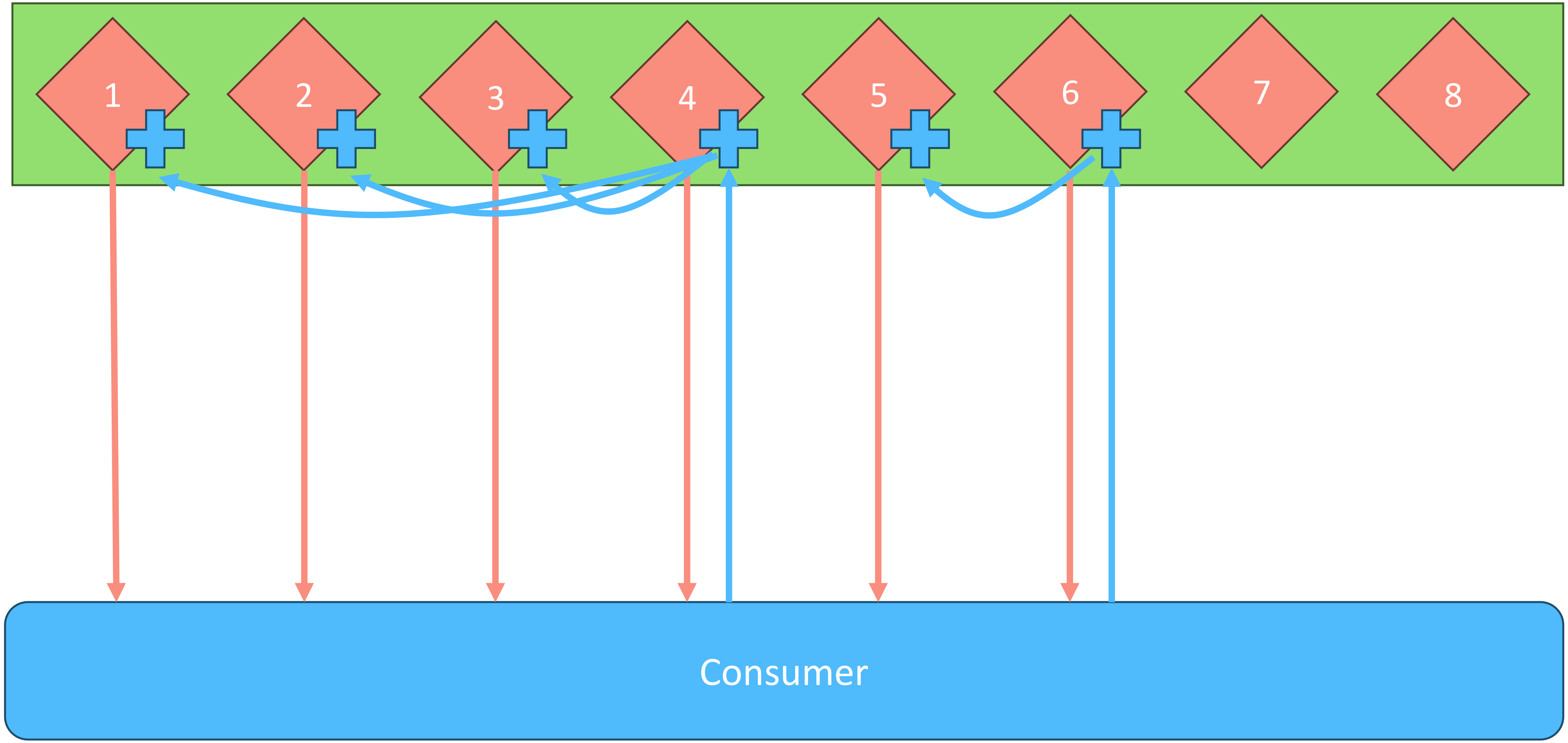

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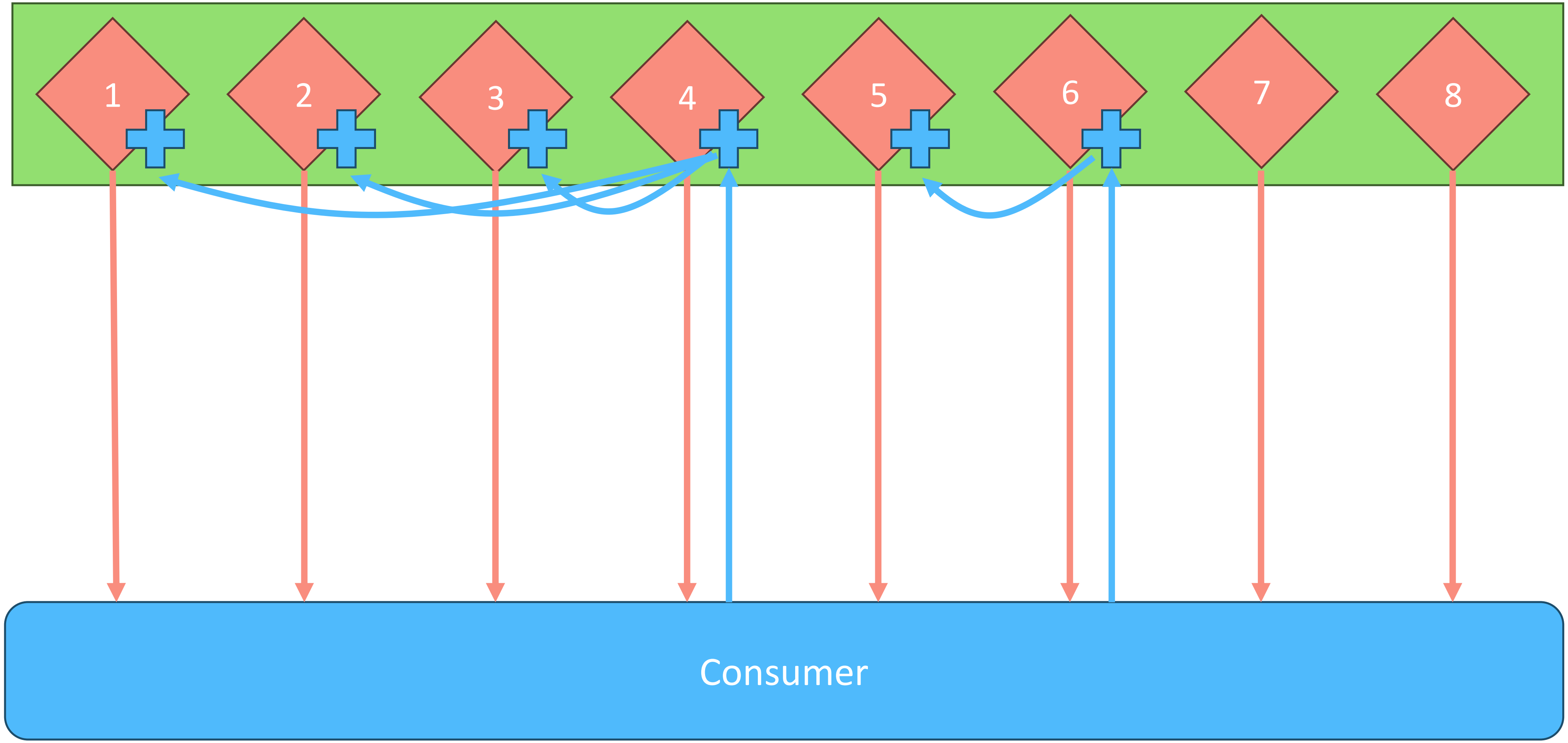

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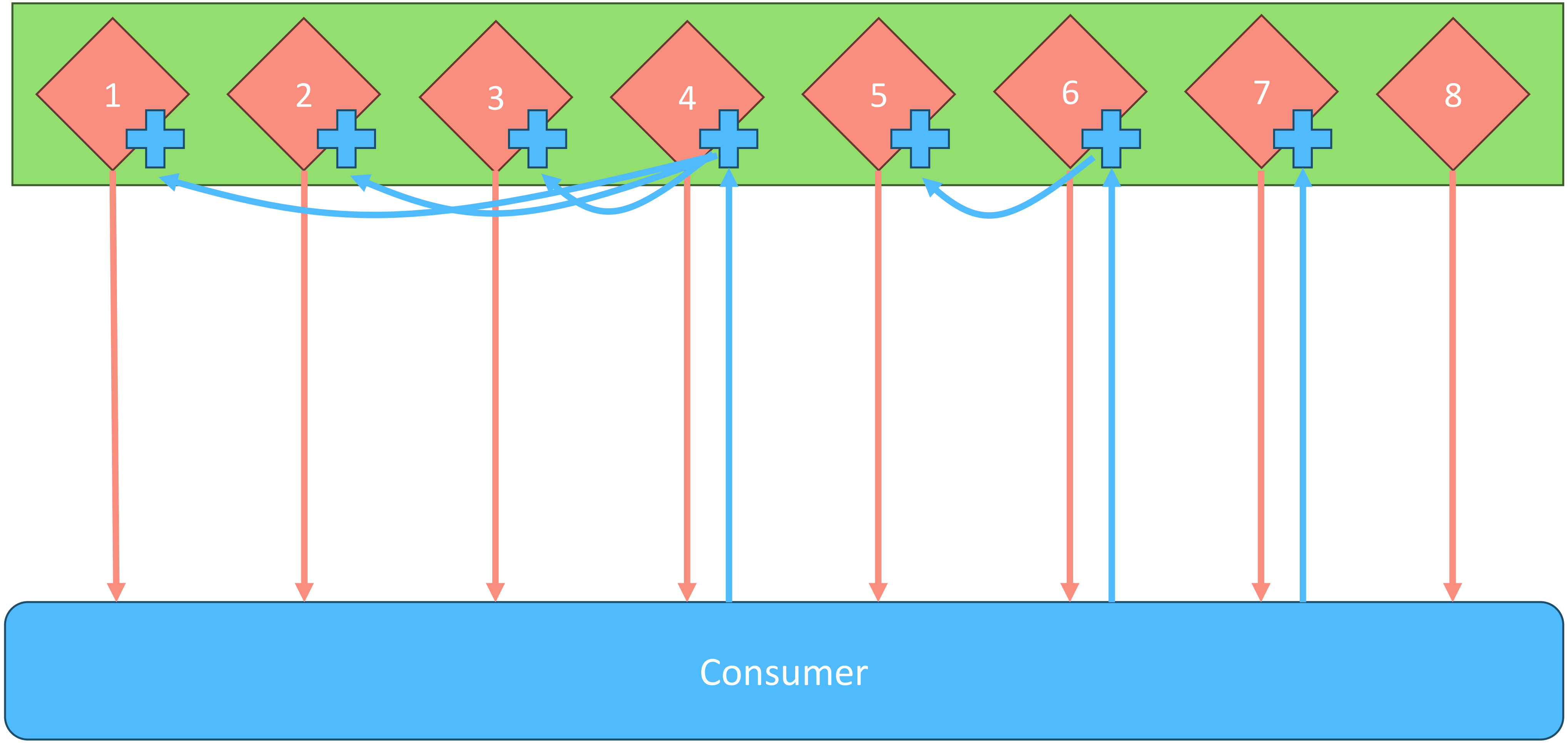

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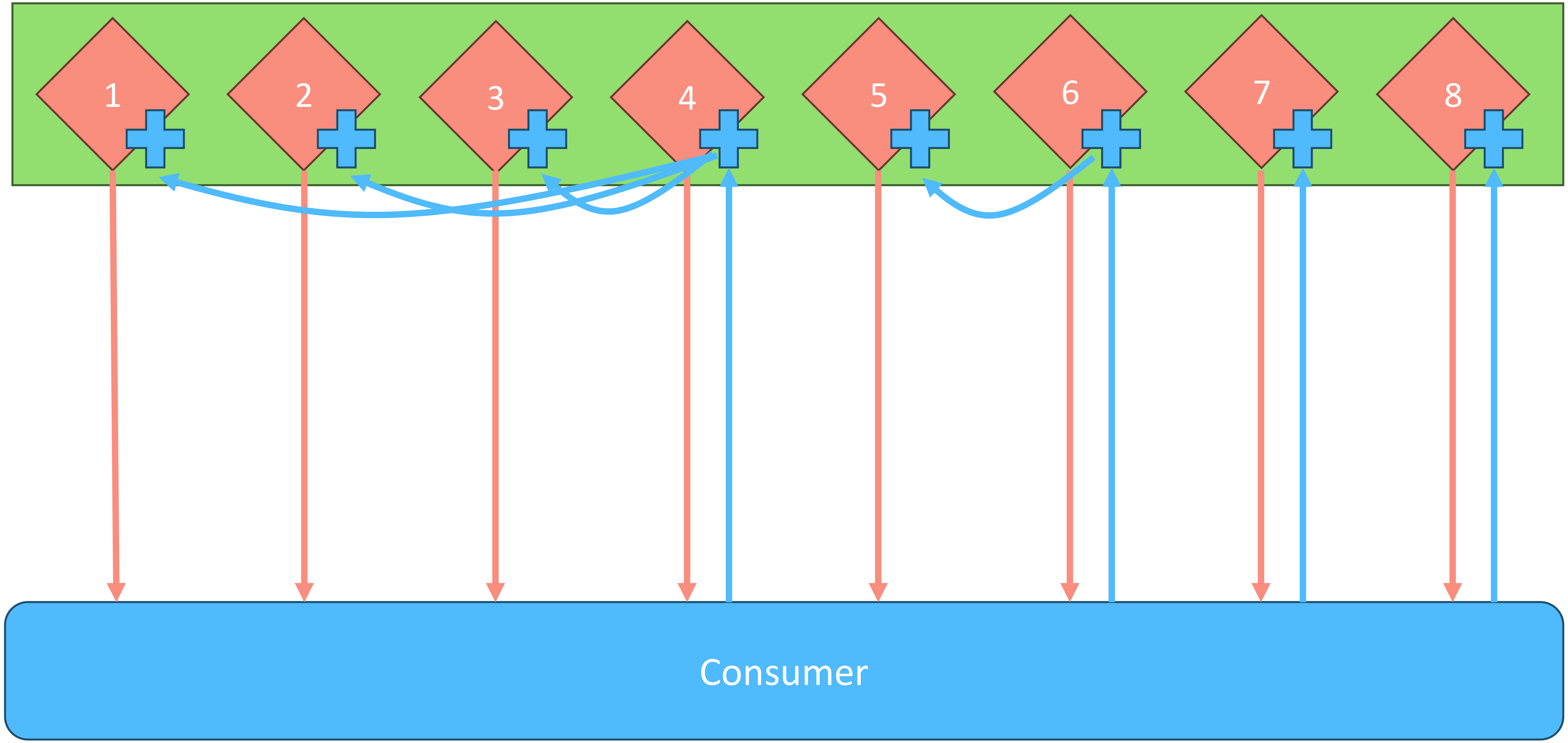

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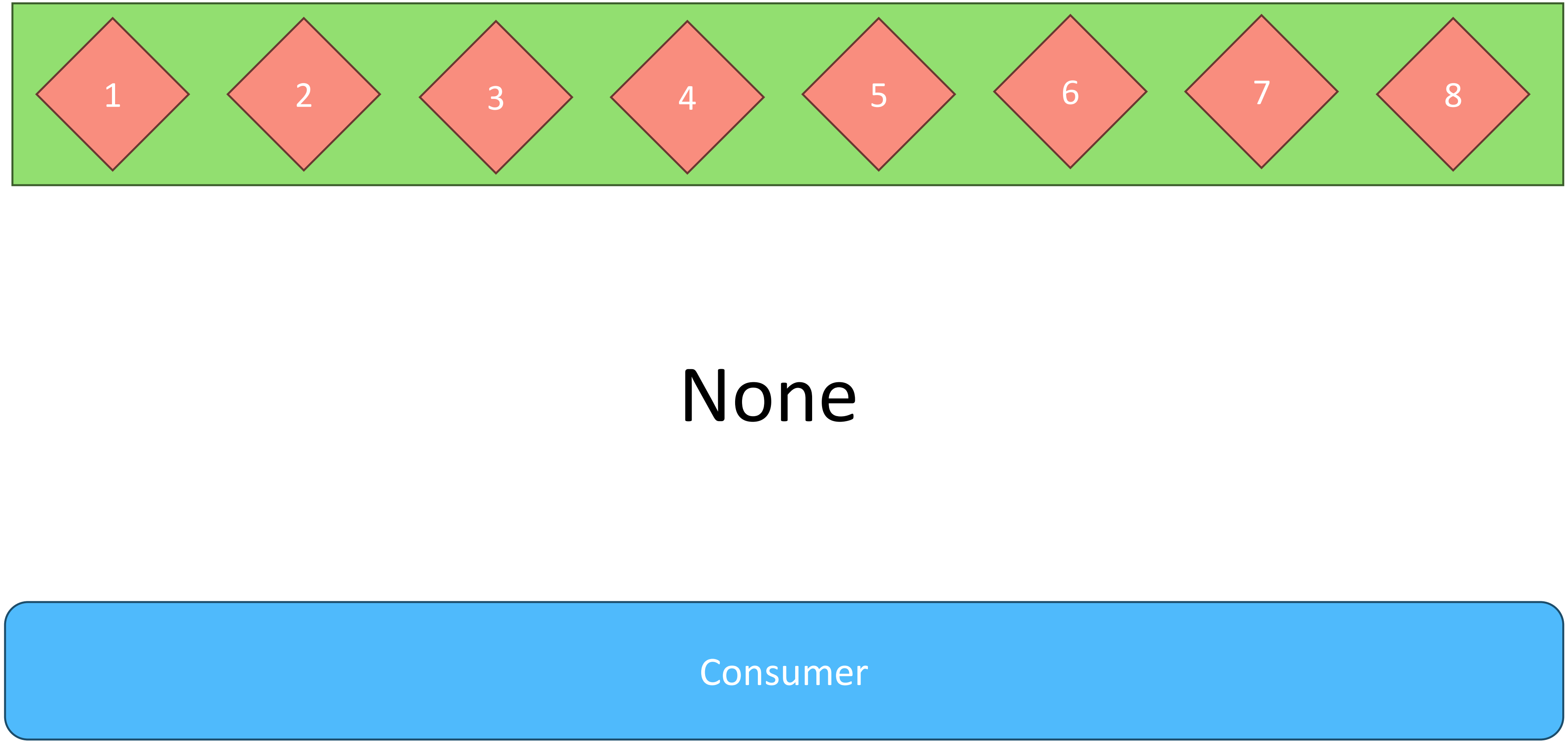

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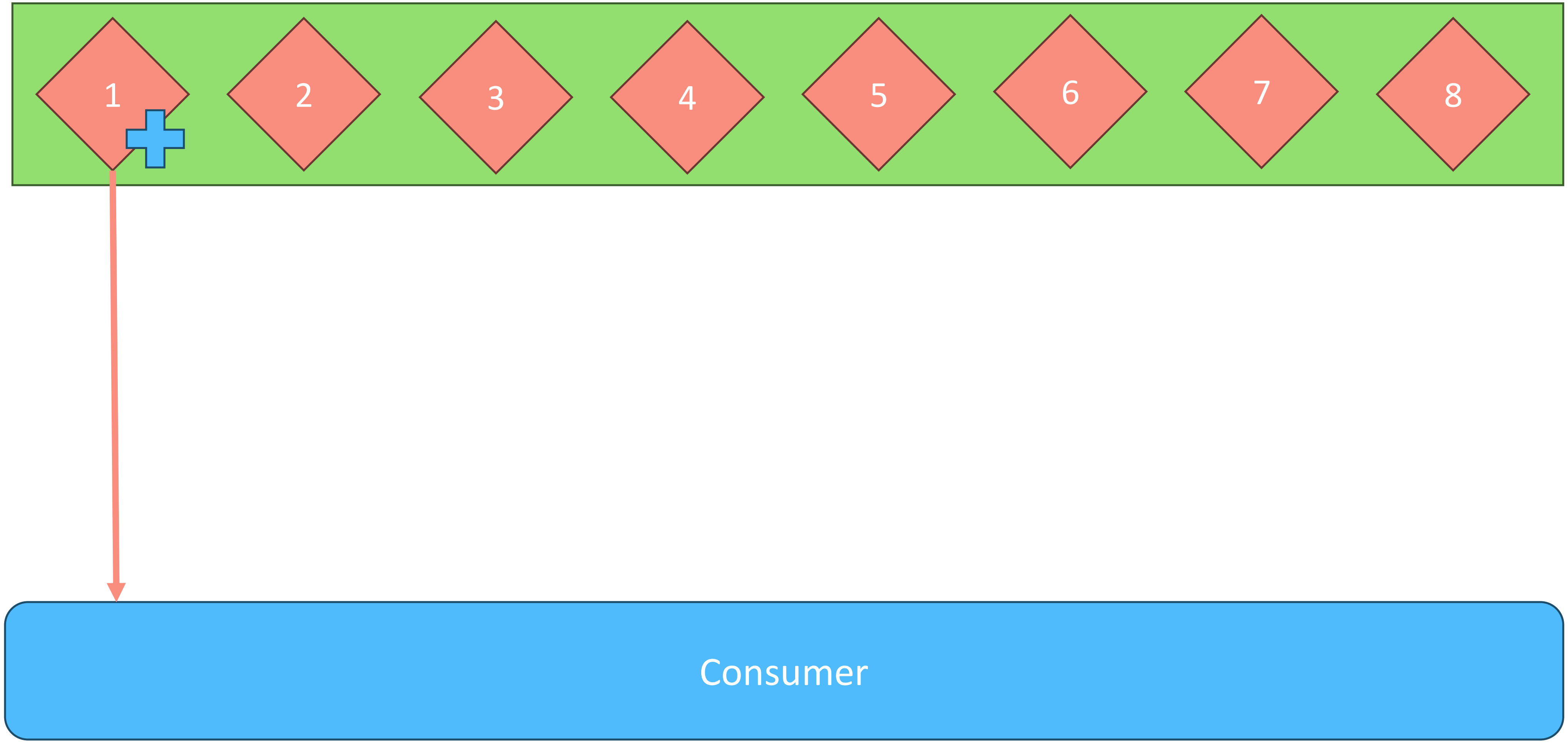

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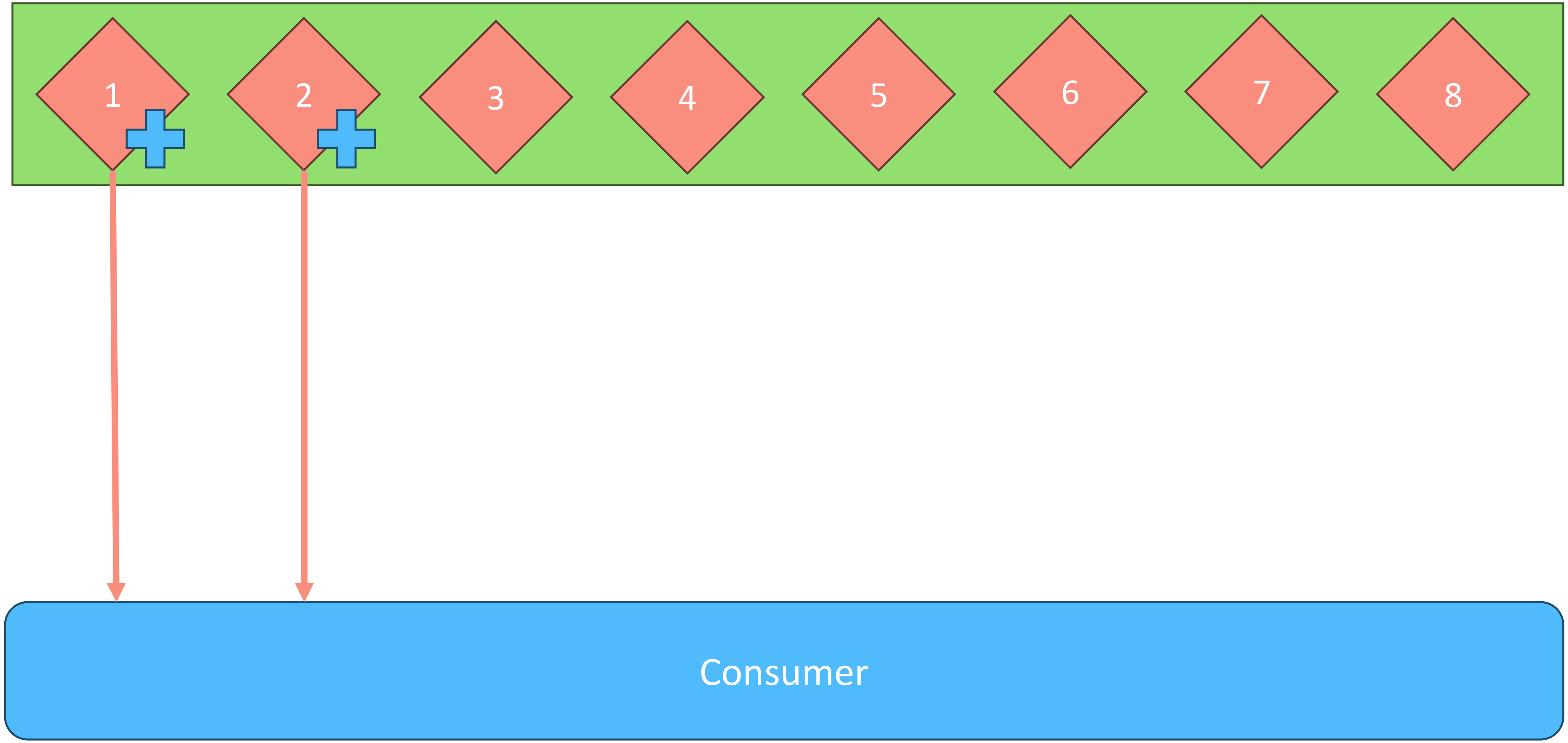

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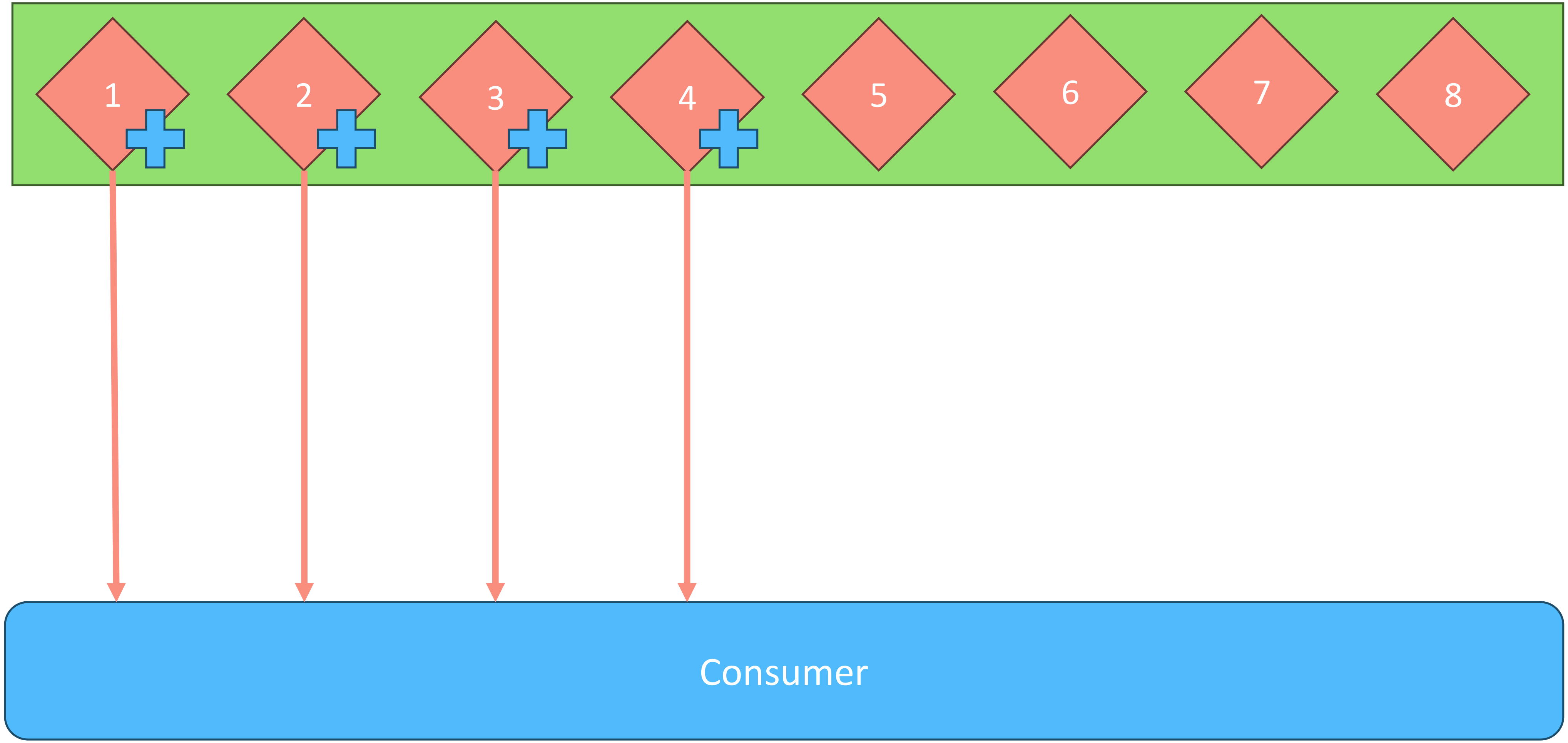

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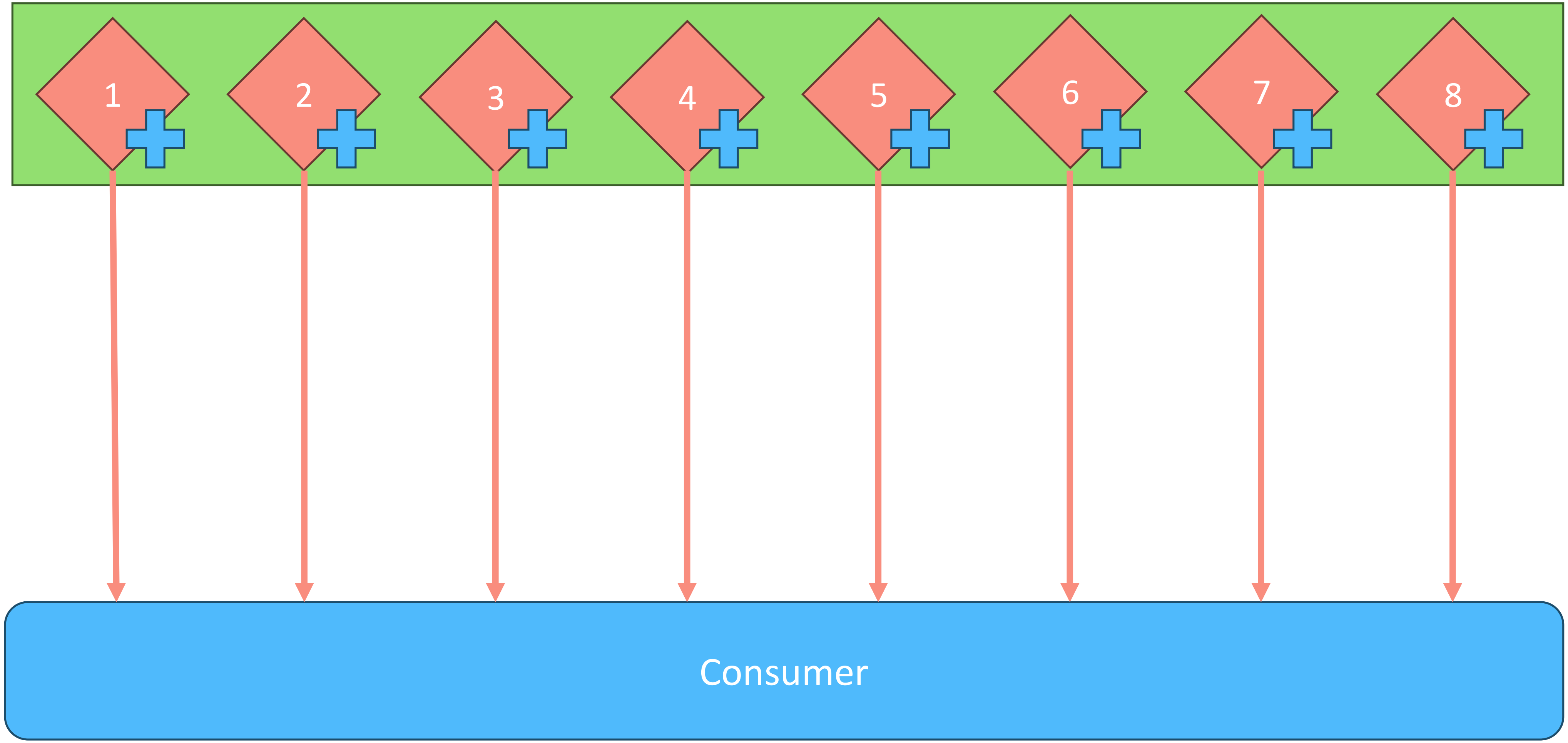
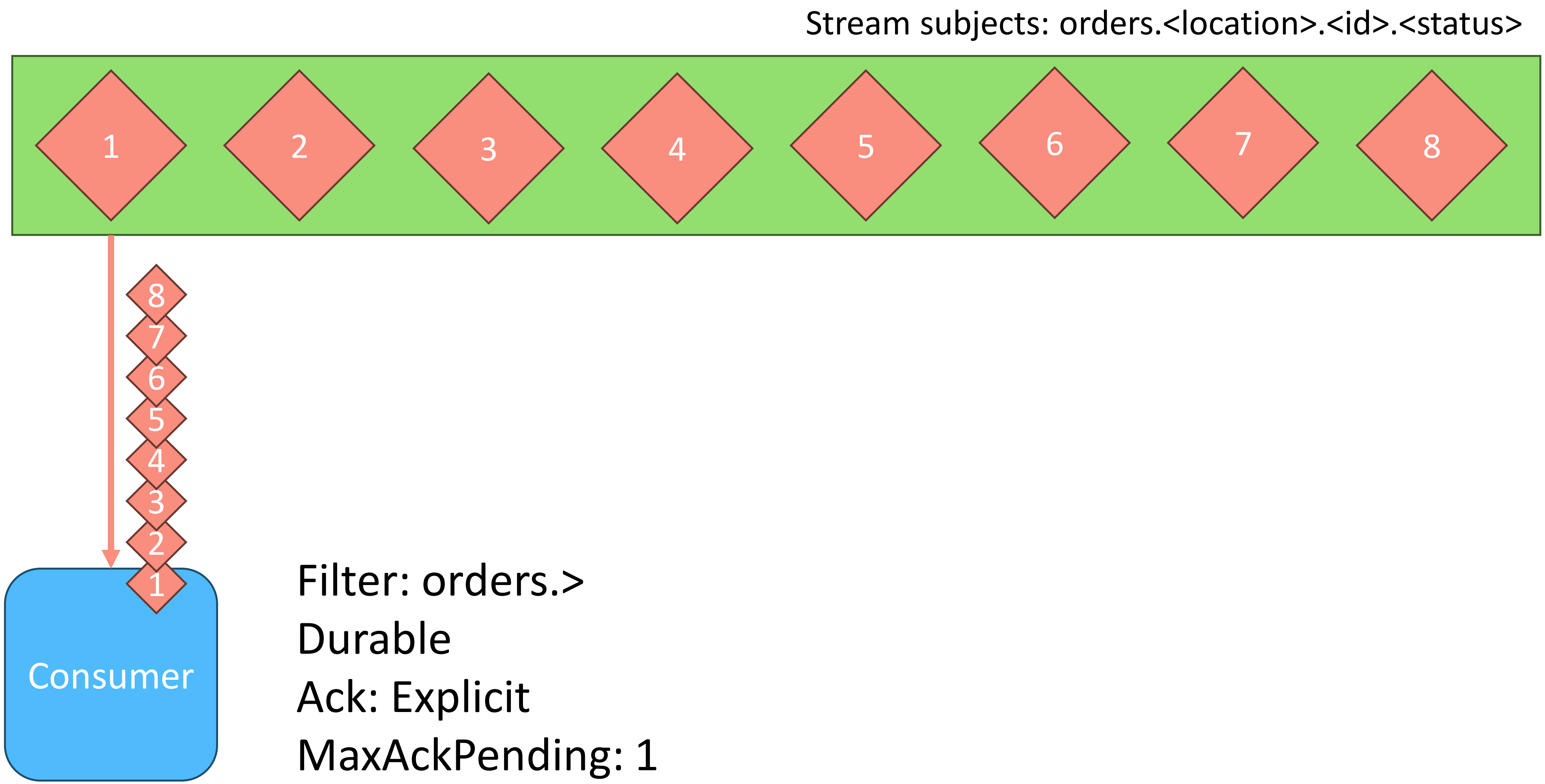


JS Consumers: Moving Parts

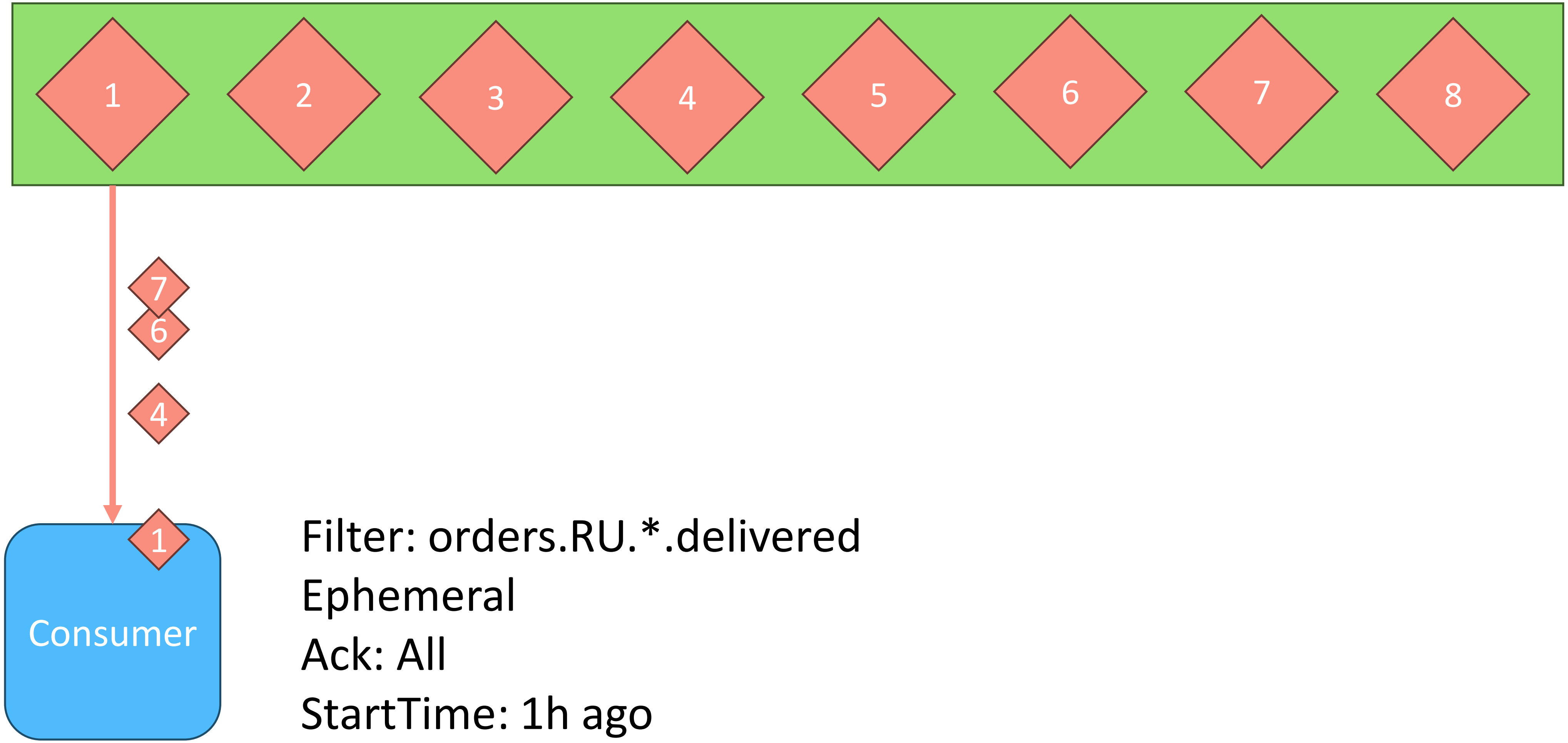
- Filtering (Shaping traffic to consumer)
- Durability (Survives restarts or reconnects)
- Acks, Ack Wait (Control redelivery behavior)
- MaxAckPending (Controls batching vs order)
- Deliver Policy (From the beginning, only recent or new)
- Replay Policy (”real-time” vs “fast”)
- Headers Only (For monitoring or filtering)

Consumer Filtering: EventSourcing



Consumer Filtering: Reporting

Stream subjects: orders.<location>.<id>.<status>



Filter: orders.RU.*.delivered
Ephemeral
Ack: All
StartTime: 1h ago

Consumer Filtering: Select/Lookup

Stream subjects: orders.<location>.<id>.<status>

Filter: orders.RU.7786542.*
Ephemeral
Ack: None
LastPerSubject

What else?

NATS: More Features

- NATS KV
- NATS Object Store
- NATS Execution Engine (Nex)
- Subject Mappings
- Virtual Streams

NATS JetStream Key/Value Store

- Distributed, versioned map (bucket) built on top of JetStream
- Designed for small configs/secrets with automatic replication & history
- Event-driven: watch subscriptions, push updates instantly across the cluster
- Ideal for: feature flags, app config, service discovery, leader election tokens

Cons / Limits:

- Value size capped (default 1 MB) – not for large blobs
- Version history limited to bucket's --history setting
- No server-side transactions across multiple keys
- Latency depends on stream replication quorum

NATS JetStream Object Store

- Content-addressed blob storage using JetStream for chunks & metadata
- Handles files of virtually any size with stream-based download/upload
- Replicated across NATS cluster; no external S3 required
- Use cases: backup snapshots, artifacts, static assets

Cons / Limits:

- Objects are immutable once stored (update = new object)
- No partial writes, entire object (re)upload
- Performs best with sequential access; random reads not optimized
- Storage/back-pressure quotas identical to underlying JetStream limits
- Not a distributed storage system. Must fit into storage fs/memory

NATS Execution Engine (Nex)

- Serverless for NATS

- Lightweight, pluggable add-on to any NATS
- Deploy “serverless” functions (JavaScript, WebAssembly) and long-running services (statically compiled Linux binaries) with zero SDKs or heavy dependencies
- Leverages existing NATS primitives (Streams, Key-Value, Object Store) for built-in reliability, security, and data replication

https://www.youtube.com/watch?v=EfxtiKMnoyQ

Subject Mappings

Simple mapping of subjects
(like map in Kafka Streams, but *inside*)

Why?

- Migrations and evolution
- Filtering
- For multi zone deployments message routing

Virtual Streams in JetStream

- Virtual stream: combination of real streams + mappings from real streams
- Consists of
  - Write streams (regular JS)
  - Read streams (mirror or source) for local regional access
  - Subject Mappings (publishing & consumption)

Why?

- Worldwide clusters
- Read from local zone without delay
- Write to local zone, then eventually replicate globally

Further reading

Further reading

- https://natsbyexample.com/
- https://docs.nats.io/nats-concepts/overview/compare-nats
- https://github.com/synadia-io/rethink_connectivity
- https://www.youtube.com/@SynadiaCommunications

- https://www.synadia.com/blog/multi-cluster-consistency-models
- https://www.youtube.com/watch?v=EfxtiKMnoyQ
- https://www.youtube.com/watch?v=334XuMma1fk

Homework

Play with NATS

- Test Pub/Sub
- Test Work Queue (group consumers)
- Test Request/Response

Demo